MASTER THESIS

Null controllability of a semilinear wave equation

Theoretical and Numerical study

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Dedicated to my inspiring parents and brothers.

Abstract

This master thesis deals with the theoretical and numerical study of the controllability problem of the semilinear wave equation, we consider interior controls that depends on time and space variable, such controls that drive the solution from a prescribed initial state to zero at a large enough controllability time. The goal is to introduce and analyze a unified approach is presented for proving quadratic convergence of the approximate solutions of the linearized semilinear wave equation with interior control in 1-D case, obtained by Newton's method and the continuation method to guarantee the adequate choice of our initial guess. Numerically, we transform the controllability problem into a mixed formulation problem developed in [Cindea and Munch, A mixed formulation for the direct approximation of the control of minimal $L2$-norm for linear type wave equations] and solved in the framework of the (space-time) finite element method, is particularly well-adapted to address the case of time dependent support. Several numerical experiments are discussed.

Keywords— Nonlinear hyperbolic PDEs, Null controllability, Newton's method, Continuation method

1 Introduction

Partial differential equations are associated to a certain good class of operators, which gives precise Mathematical models to real phenomena : Physics (conservation laws, quantum mechanics), mechanics (fluid mechanics, elasticity), economics (macroeconomic models, stock management) and other scientific fields. The interests of modeling a physical phenomenon using mathematical objects is to understand the phenomenon and the influence of its different parameters, and as a consequence the study of all the possible outcomes using simulations. The main goal of control theory in general is the possibility of acting on certain system using a control in an optimum manner, or with less cost without delay or overshoot and ensuring control stability.

The controllability problem is a particular problem in control theory and it may be formulated as follows : consider an evolution problem (either described in terms of Partial or Ordinary Differential Equations).We are allowed to act on the trajectories of the system by means of a suitable choice of the control (the right hand side of the system, the boundary conditions, etc.).Then, given a time interval $t(0,T)$, and initial and final states we have to find a control such that the solution matches both the initial state at time $t=0$ and the final one at time $t=T$.
-This is a classical problem in Control Theory and there is a large literature on the topic : The book of Lee and Markus [8] for finite dimensional systems ,and J-L Lions [7] for an introduction of the controllability problem of partial differential equaitons.
This paper is devoted to the controllability of a semilinear wave equation : Let $\Omega=(0,1)$, $\omega=(l_1,l_2)\subset\Omega$ and denote $Q_T=(0,T)\times\Omega$. Let $f:\mathbb{R}\to\mathbb{R}$ a $C^1(\mathbb{R})$. GivenT $T>0$ consider the following non-homogeneous wave equation :

$$\begin{cases} y_{tt}-\Delta y+f(y)=h1_\omega & (t,x)\in Q_T,\\ y(.,0)=y(.,1)=0 & t\in(0,T),\\ y(0,.)=y^0,\ y_t(0,.)=y^1 & x\in\Omega, \end{cases} \tag{.1}$$

where $(y^0,y^1)\in H_0^1(\Omega)\times L^2(\Omega):=\mathcal{V}$ is the initial data and $h\in L^2(q_T)$ where $q_T:=(0,T)\times\omega$ is an interior control function.
The exact controllability problem is formulated as follows : for any $(y^0,y^1),(z^0,z^1)\in\mathcal{V}$, find a control function $h\in L^2(q_T)$ such that the state y solution of (.1) satisfies $(y(T,.),y_t(T,.))=(z^0,z^1)$ at time $T>0$.
Assuming the following growth condition on the nonlinearity f : there is $\beta_0>0$ such that

$$\lim_{|s|\to+\infty}\frac{|f(s)|}{|s|\log^2|s|}<\beta_0, \tag{.2}$$

the controllability problem is solved in [1] for large $T>0$:

Theorem 1.1. *[1] Assume $T>2\max(l_1,1-l_2)$. There exists a $\beta_0>0$ such that if the nonlinearity $f\in C^1(\mathbb{R})$ satifies* (.2)*, then for all $(y^0,y^1)\in\mathcal{V}$ there exists controls $h\in L^2(q_T)$ such that the solution of* (.1) *satifies $(y(T,.),y_t(T,.))=(z^0,z^1)$ for all $(z^0,z^1)\in\mathcal{V}$.*

The growth condition (.2) is almost optimal in the sense that if f behaves at infinity like $-s\log^p|s|$ with $p>2$, then due to the bow-up phenomena, the system is not exactly controllable in any time $T>0$.
Section 2 is devoted to the study of the following abstract controllability problem

$$\begin{cases} \frac{d}{dt}y=Ay+Bh,\\ y(0)=y^0\in H, \end{cases} \tag{.3}$$

where we define some notions in spectral theory which allows us to give existence and uniqueness results for the weak solution of the linear wave equation using the Duhamel formula. We give

the general concepts of controllability by defining the main notions : Exact, Null and Approximate controllability, we also study the difference between these notions in a infinite and finite dimensional case where all these notions coincide and are equivalent to the Kalman Rank Condition [2], but in an infinite dimensional case we only have the equivalence between exact and null controllability under certain hypothesis. Then we move to the notions of observability, and we prove the observability inequality in a special case and as a consequence we will deduce the exact controllability of the system. Section 3 deals with the particular case : $A = \begin{pmatrix} 0 & 0 \\ -\Delta + \xi & 0 \end{pmatrix}$, and $B = 1_\omega$ that defines the wave operator. First of all we give equivalent definitions of exact, null, and approximate controllability involving the notion of *Reachable set* $R(T;(y^0,y^1))$ We consider the following linear wave equation with a potential ξ where $\xi \in L^\infty(Q_T)$, and a control $h \in L^2(q_T)$

$$\begin{cases} y_{tt} - \Delta y + \xi y = h 1_\omega & (t,x) \in Q_T, \\ y(t,0) = y(t,1) = 0 & t \in (0,T), \\ y(0,x) = y^0(x) \in H_0^1(\Omega) \quad y_t(0,x) = y^1(x) \in L^2(\Omega) & x \in \Omega. \end{cases} \tag{.4}$$

And in order to obtain the minimal L^2 norm control, which is the solution of (.5), we define a functional $\mathcal{J}$ on an adequate Hilbert $\mathcal{H}$ space and by minimizing $\mathcal{J}$ we obtain the initial condition $(\phi^0, \phi^1) \in \mathcal{H}$ of the following adjoint sysem

$$\begin{cases} \phi_{tt} - \Delta \phi + \xi \phi = 0 & (t,x) \in Q_T, \\ \phi(t,0) = \phi(t,1) = 0 & t \in (0,T), \\ \phi(0,x) = \phi^0(x) \quad \phi_t(0,x) = \phi^1(x) & x \in \Omega, \end{cases} \tag{.5}$$

and finally, we prove the observability of the adjoint system (.5) for all $\xi \in L^\infty(Q_T)$

$$||(\phi^0, \phi^1)||_{\mathcal{H}}^2 \leq A_1 e^{B_1 \sqrt{||\xi||_\infty}} \int_{q_T} \phi^2 dt dx, \tag{.6}$$

for some $A_1, B_1 > 0$, and as a consequence the exact controllability of (.4).
Section 4 is devoted to the operator nonlinear operator K defined in [1] based on the HUM method, and based on the observability property as in [1] we deduce the existence of a fixed point of the mapping K, hence a solution to the controlability of the semilinear wave equation through the control of minimal L^2 norm. We prove mainly in which case the operator K is a contraction considering the perturbed system of the linear wave equation.
Finally sections 5 and 6 deal with the numerical study of the semiliner wave equation, where we define the following Newton operator

$$\begin{array}{rccl} F: & [0,1] \times X & \mapsto & \mathcal{X} \\ & (\lambda, U) & \to & \big(Ly + \lambda f(y) - \phi 1_\omega, (y(0,.) - y^0, y_t(0,.) - y^1), (y(T,.), y_t(T,.))\big), \end{array}$$

with adequate Hilbert spaces X and $\mathcal{X}$.

2 Controllability of partial differential equation

This section is devoted to the controllability of partial differential equation in the general case. The various types of controllability are introduced and characterized using the duality approach, where we prove that exact controllability is equivalent to the observability of the adjoint problem.
Let Ω and $\omega \subset \Omega$ two open sets of $\mathbb{R}$, we consider the following control system where H is a suiatable Hilbert space

$$\begin{cases} P(D)y = h1_\omega & t > 0,\ x \in \Omega, \\ B(D)y = 0 & t > 0,\ x \in \partial\Omega, \\ y(0,.) = z^0 \in H & x \in \Omega, \end{cases} \tag{.1}$$

where $P(D)$ is the differential operator ($P \in \mathbb{C}(\tau, \xi)$ and $D = (-i\partial_t, -i\partial_x)$). We take in the special case of the linear wave equation $P = -\tau^2 + |\xi|^2$.
here $h = h(t,x)$ is the interior control, $z = z(t,x)$ is the state. And the controllability problem may be formulated as follows : we are allowed to act on the trajectories of the system by means of a suitable choice of the control such that the solution of (.1) satifies :

$$\forall z^0, z^1 \in H,\ \ z(T,.) = z^1.$$

Firstly, we are interested with some results of spectral theory which answers the existence and uniqueness problem of solutions of an abstract problem of controllability.

Unbounded operators

Definition 2.1. *Let H be a separable Hilbert space on $\mathbb{R}$, an unbounded operator on H is a pair $(D(A), A)$ such that :*
$A : D(A) \to H$ is a linear operator.
The space $D(A) \subset H$ is the domain of the unbounded operator.

Example 2.1. *Let $H = L^2(\mathbb{R}^n)$, we have the following unbounded operators : $(\mathcal{C}_0^\infty(\mathbb{R}^n), -\Delta)$ and $(\mathcal{C}^2(\mathbb{R}^n), -\Delta)$ or even $(H^2(\mathbb{R}^n), -\Delta)$.*

Of course these operators are related, the following definition meant to clarify this situation.

Definition 2.2. *If $D \subset D'$ and $Au = A'u$ for all $u \in D'$, we say that (D, A) is an extension of (D', A'), and we denote $(D', A') \subset (D, A)$.*

Example 2.2. *For example we have*

$$(\mathcal{C}_0^\infty(\mathbb{R}^n), -\Delta) \subset (\mathcal{C}^2(\mathbb{R}^n), -\Delta) \subset (H^2(\mathbb{R}^n), -\Delta).$$

Definition 2.3. *An unbounded operator $(D(A), A)$ is bounded when the quantity*

$$||A|| = \sup\{||Au||_H,\ u \in D(A),\ ||u||_H = 1\},$$

is finite.

Definition 2.4. *Let $(D(A), A)$ be an unbounded operator, and $G = \{(u, Au)_H,\ u \in D(A)\}$ its graph, we say that $(D(A), A)$ is closed when G is a closed subset of $H \times H$.*

Proposition 2.1. *An unbounded operator $(D(A), A)$ on H is bounded if and only if $D(A) = H$ and $(D(A), A)$ is closed.*

Adjoint of unbounded operator

Given $(D(A), A)$ an unbounded operator on H, with dense domain. Let $x \in H$, if we suppose that it exists $y \in H$ such that for all $z \in D(A)$, $(x, Az)_H = (y, z)_H$. Then y is unique. Indeed if $(y, z)_H = 0$ for all $z \in D(A)$, then $y \in D(A)^\perp = \{0\}$.
The set $\{x \in H,\ \exists! y,\ (x, Az)_H = (y, z)_H\ \forall z \in H\}$ for which is denoted $D(A^*)$, we denote $A^*x = y$. This defined the adjoint of $(D(A), A)$.

Definition 2.5. *Let $(D(A), A)$ be an unbounded operator on H, with dense domain. The adjoint of $(D(A), A)$ is the unbounded operator $(D(A^*)A^*)$ given by*

$$\begin{cases} D(A^*) = \Big\{x \in H,\ \exists y \in D(A),\ \textit{for all } z \in D(A),\ (x, Az) = (y, z)\Big\}, \\ \textit{for } x \in D(A^*),\ A^*x = y. \end{cases}$$

Example 2.3. *The adjoint of $(D(A), A) = (\mathcal{C}_0^\infty(\mathbb{R}^n), -\Delta)$ is $(H^2(\mathbb{R}^n), -\Delta)$.*
Indeed, for $f \in L^2(\mathbb{R}^n)$, and $\phi \in \mathcal{C}_0^\infty(\mathbb{R}^n)$ we have :

$$(f, -\Delta)_{L^2(\mathbb{R}^n)} = (\hat{f}, \widehat{-\Delta\phi})_{L^2(\mathbb{R}^n)} = \int_{\mathbb{R}^n} |s|^2 \hat{f}(s)\hat{\phi}(s)ds,$$

thus there exists $g \in L^2(\mathbb{R}^n)$ such that $(\hat{f}, \widehat{-\Delta\phi})_{L^2(\mathbb{R}^n)} = (g, \phi)_{L^2(\mathbb{R}^n)} = (\hat{g}, \hat{\phi})_{L^2(\mathbb{R}^n)}$ for all $\phi \in \mathcal{C}_0^\infty(\mathbb{R}^n)$ if and only if $s \mapsto |s|^2\hat{f}(s)$ belongs to $L^2(\mathbb{R}^n)$, which is equivalent to $f \in H^2(\mathbb{R}^n)$ (since $f \in L^2(\mathbb{R}^n)$).
Hence $A^*f = g = -\Delta f$.

Definition 2.6. *An unbounded operator $(D(A), A)$ is symmetric when $(D(A), A) \subset (D(A^*), A^*)$, and*

$$(Ax, y)_H = (x, Ay)_H, \quad \forall x, y \in D(A).$$

Definition 2.7. *An unbounded operator with dense domain $(D(A), A)$ is self-adjoint (resp. skew-adjoint) when $(D(A^*), A^*) = (D(A), A)$ (resp. $(D(A^*), A^*) = (D(A), -A)$).*

Example 2.4. *If we consider the wave operator on $X = H_0^1(\Omega) \times L^2(\Omega)$*

$$\begin{cases} D(A) = \Big\{(u, v) \in X,\ \Delta u \in L^2(\Omega),\ v \in H_0^1(\Omega)\Big\}, \\ A(u, v) = (v, \Delta u),\ \ \forall (u, v) \in D(A). \end{cases}$$

A is skew-adjoint, indeed, we have for all $(u, v), (w, z) \in D(A)$ we have

$$\begin{array}{rcl} (A(u, v), (w, z))_X & = & \int_\Omega [\nabla v.\nabla w + \Delta u z]dx \\ & = & -\int_\Omega [\nabla u.\nabla z + \Delta w v]dx \\ & = & ((u, v), (-A)(w, z))_X. \end{array}$$

Definition 2.8. *A family of bounded operators $(S(t))_{t\geq 0}$ defined on H is called a continuous semigroup is the following assumptions are true :*
(i) $S(0) = I$,
(ii) $S(t + s) = S(t)S(s)\ \forall t, s \geq 0$,
(iii) $S(t) \in C^0(H, H)\ \forall t \geq 0$,
(iv) $\forall u \in H,\ t \mapsto S(t)u \in C^0((0, +\infty), H)$.

Remark 2.1. *If the mapping $S(t)$ from H into H are defined for $t \in \mathbb{R}$, if the properties (ii), (iii) hold for any $t, s \in \mathbb{R}$ and if, in (iv), $(0, +\infty)$ can be replaced by $\mathbb{R}$, then $(S(t))_{t\in\mathbb{R}}$ is a continuous group.*

Definition 2.9. *A family of bounded operators $(S(t))_{t\geq 0}$ defined on H is called a strongly continuous semigroup on H if the following assumptions are true :*
(i) $S(0) = I$,
(ii) $S(t + s) = S(t)S(s)\ \forall t, s \geq 0$,
(iii) $\lim\limits_{t\to 0} ||S(t)x - x||_H = 0\ \forall x \in H$.

Definition 2.10. *Let $(S(t))_{t\geq 0}$ be a semigroup strongly continuous on H. We define the infinitesimal generator of the semigroup $(S(t))_{t\geq 0}$, the unbounded operator $(D(A), A)$ defined as follows*

$$\begin{cases} D(A) = \left\{x \in H,\ \lim_{t\to 0} \frac{S(t)x-x}{t} \in H\right\}, \\ Ax = \lim\limits_{t\to 0} \frac{S(t)x-x}{t}. \end{cases}$$

Concepts of controllability

Let $A : D(A) \subset H \to H$ is an unbounded operator incorporating the differential operator together with the homogeneous boundary conditions, and $B : U \to H$ is a bounded operator representing the effect of the control h. Here H and U represents appropriate real Hilbert spaces of functions. We shall also assume that A generates a strongly continuous semigroup of operators on H, denoted by $(S(t))_{t\geq 0}$. We are interested in the following linear problem :

$$\frac{d}{dt}y = Ay + Bh.$$

Example 2.5. *In the interior control problem of the linear wave equation :*

$$\begin{cases} y_{tt} - \Delta y = h1_\omega & (t,x) \in Q_T, \\ y(t,0) = y(t,1) = 0 & t \in (0,T), \\ y(0,x) = y^0(x) \quad y_t(0,x) = y^1(x) & x \in \Omega, \end{cases}$$

may be written as follows

$$\frac{d}{dt}y = Ay + Bh.$$

and we define the linear operator A on $H = H_0^1(\Omega) \times L^2(\Omega)$ by

$$\begin{cases} D(A) = \left\{(u,v) \in H,\ \Delta u \in L^2(\Omega),\ v \in H_0^1(\Omega)\right\}, \\ A(u,v) = (v, \Delta u),\ \forall (u,v) \in D(A). \end{cases}$$

$Bh = h1_\omega$ and $y(0) = (y^0, y^1)$.

Theorem 2.1. *Assume that A generates a strongly continuous semigroup of operators on H, denoted by $(S(t))_{t\geq 0}$ (or $(e^{tA})_{t\geq 0}$). Let $y^0 \in H$ and $h \in L^2(0,T;U)$ the following Cauchy problem*

$$\begin{cases} \frac{d}{dt}y = Ay + Bh, \\ y(0) = y^0 \in H, \end{cases}$$

admits a unique weak solution $y \in C([0,T]; D(A)) \cap C^1([0,T]; H)$ given by the Duhamel formula :

$$y(t) = S(t)y^0 + \int_0^t S(t-s)Bh(s)ds.$$

Now we define the main notions we consider in this document, namely the exact and Null controllability of the following system

$$\begin{cases} \frac{d}{dt}y = Ay + Bh, \\ y(0) = y^0 \in H. \end{cases} \tag{.2}$$

Definition 2.11. *The system (.2) is **exactly controllable in time T** if and only if for every $y^0, y^T \in H$ there exists $h \in L^2(0,T;U)$ such that :*

$$y(T) = y^T.$$

Definition 2.12. *The system* (.2) *is* ***Null Controllable in time T*** *if and only if for every* $y^0 \in H$ *there exists* $h \in L^2(0,T;U)$ *such that :*

$$y(T) = 0.$$

Definition 2.13. *The system* (.2) *is* ***Approximately Controllable in time T*** *if and only if for every* $\epsilon > 0$ *and* $y^0, y^T \in H$ *there exists* $h \in L^2(0,T;U)$ *such that :*

$$||y(T) - y^T|| < \epsilon.$$

Let us introduce the operator $L_T : L^2(0,T;U) \to H$

$$L_T h = \int_0^T S(T-s)Bhds.$$

Then we have the following result :

Proposition 2.2.

$$\begin{array}{lll} (1)\ \textit{system (.2) is Exactly controllable in time T} & \Longleftrightarrow & \mathsf{Im}L_T = H, \\ (2)\ \textit{system (.2) is Null controllable in time T} & \Longleftrightarrow & S(T)H \subset \mathsf{Im}L_T, \\ (3)\ \textit{system (.2) is Approximatly controllable in time T} & \Longleftrightarrow & \overline{\mathsf{Im}L_T} = H. \end{array}$$

Remark 2.2. *[2]*
In finite dimension ($A \in \mathcal{M}_{n,n}$ and $B \in \mathcal{M}_{n,m}$), these three notions are equivalent, and equivalent to the Kalman Rank Condition :

$$\textit{rank}(B, AB, ..., A^{n-1}B) = n.$$

Remark 2.3. *In infinite dimensional spaces these notions are no longer equivalent, but we have then following implications :*

$$\begin{array}{lll} \textit{Exact controllability} & \Longrightarrow & \textit{Null contrabillity}, \\ \textit{Exact controllability} & \Longrightarrow & \textit{Approximate controllability.} \end{array}$$

Proposition 2.3. *If A generates a continuous group* $(S(t))_{t\in\mathbb{R}}$ *then*

$$\textit{Exact controllability} \Longleftrightarrow \textit{Null contrabillity.}$$

Proof: $''\Longleftarrow''$ Assuming the Null controllability of the following system

$$\begin{cases} \frac{d}{dt}y = Ay + Bh, \\ y(0) = y^0 \in H. \end{cases}$$

In order to prove the exact controllability of this system : Let $y^T \in H$, is there a control $h \in L^2(0,T;U)$ such that

$$y(T) = y^T,$$

Since, the system is Null controllable : there exists a control $h^0 \in L^2(0,T;U)$ such that we can drive the system from $S(-T)y^T \in H$ to zero, using Duhamel's formula :

$$y(T) = S(T)[S(-T)y^T] + \int_0^T S(T-s)Bh^0 ds.$$

In fact, there exists a control $u \in L^2(0,T;U)$ such that we can drive this system from $S(-T)y^T$ to zero :

$$0 = S(T)S(-T)y^T + \int_0^T S(T-s)Bh^0(s)ds$$

Using the semigroup property $S(T)S(-T) = S(0) = Id_H$,

$$y^T = \int_0^T S(T-s)B(-h^0(s))ds.$$

As a result, the control $h = -h^0 \in L^2(0,T;U)$ drives the system from y^0 to y^T in time T.
Hence the system is exact controllable in time T. □

Observability inequality

The following notions : *Observability inequality* will play a fundamental role in solving control problems. As we will prove that Exact controllability is equivalent to the observability inequality. We consider the boundary initial value problem

$$\begin{cases} \frac{d}{dt}y = Ay + Bh, \\ y(0) = y^0 \in H, \end{cases}$$

and its adjoint problem

$$\begin{cases} \frac{d}{dt}\phi = -A^*\phi, \\ \phi(T) = \phi^T \in H. \end{cases}$$

Note that we can solve the adjoint problem backwards in time, and we will obtain

$$\phi(t) = S^*(T-t)\phi^T.$$

We have for all $\phi^T \in H$ and $h \in L^2(0,T;U)$ the following identity :

$$(y(T), \phi^T)_H = \int_0^T (h, B^*\phi)_U dt.$$

Indeed, let $u \in C^1([0,T],U)$ and $\phi^T \in D(A^*)$, as a consequence $y, \phi \in C^1([0,T],H)$, therefore we integrate by parts

$$\begin{array}{rcl} 0 & = & \int_0^T (\frac{d}{dt}z - Ay - Bu, \phi)_H dt \\ & = & -\int_0^T (y, \frac{d}{dt}\phi + A^*\phi)_H dt + [(y,\phi)_H]_0^T - \int_0^T (u, B^*\phi)_U dt \\ & = & (y(T), \phi^T)_H - \int_0^T (u, B^*\phi)_U dt. \end{array}$$

Hence we deduce from density theorems that for all $u \in L^2(0,T;U)$ and $\phi^T \in H$:

$$(y(T), \phi^T)_H = \int_0^T (h, B^*\phi)_U dt.$$

Theorem 2.2. *The system is exactly controllable in time $T > 0$ if and only if there exists $c > 0$ such that :*

$$||\phi^0||_H^2 \leq c \int_0^T ||B^*S^*(t)\phi^0||_U^2 dt, \ \ \forall \phi^0 \in H.$$

The observability inequality guarantees that the solution of the adjoint problem at $t = 0$ is uniquely determined by the observed quantity $B^*S^*(t)\phi^0$ for $0 < t < T$.

Proof: Assuming exact controllability of the system, then the map L_T is onto.
Let $\gamma : H \to L^2(0,T;U)$ denote the inverse of the restriction of L_T to $Ker(L_T)^\perp$, so that $h = \gamma(y^T)$ drives the system from 0 to y^T. Let $\phi^T \in H$ and take $y^T = \phi^T$, $h = \gamma(y^T)$, we obtain

$$\begin{array}{rcl} ||\phi^T||_H^2 & = & (y(T), \phi^T)_H \\ & = & \int_0^T (\gamma(y^T), B^*\phi)_U dt \\ & \leq & ||\gamma||.||\phi^T||_H.\big(\int_0^T ||B^*\phi||_U^2 dt\big)^{\frac{1}{2}}. \end{array}$$

reciprocally, assume that there is $c > 0$:

$$||\phi^T||_H^2 \leq c \int_0^T ||B^*\phi||_U^2 dt$$

For any $\phi^T \in H$, we set $h = B^*\phi$ where ϕ is the solution of the adjoint system, and y the solution of the system with the control h.
This defines a bounded operator $\Psi : \phi^T \in H \mapsto y(T) = L_T(B^*\phi) \in H$, indeed,

$$(\Psi\phi^T, \phi^T)_H = \int_0^T ||B^*\phi(t)||_U^2 dt \geq c||\phi^T||_H^2,$$

It follows from Lax-Milgram theorem that Ψ is invertible. Hence, for any $y^T \in H$, the state $h^T = \Psi^{-1}(z^T)$ is such that the control $h = B^*\phi$ drives the system from 0 to y^T. □

There exists a weak version of the observability inequality, which is the following

Theorem 2.3. *The system is Null controllable in time $T > 0$ if and only if there exists $c > 0$ such that :*

$$||S^*(T)\phi^0||_H^2 \leq c \int_0^T ||B^*S^*(t)\phi^0||_U^2 dt, \ \ \forall \phi^0 \in H.$$

In this case, only $S^*(T)\phi^0$ may be recovered, not ϕ^0.

3 Linear wave equation with interior control

In this section, we take the particular case : $A = \begin{pmatrix} 0 & 0 \\ -\Delta + \xi & 0 \end{pmatrix}$, and $B = 1_\omega$ that defines the wave operator. First of all we give equivalent definitions of exact, null, and approximate controllability involving the *Reachable set* $R(T;(y^0,y^1))$ We consider the following linear wave equation with a potential ξ where $\xi \in L^\infty(Q_T)$:

$$\begin{cases} y_{tt} - y_{xx} + \xi y = h1_\omega & (t,x) \in Q_T, \\ y(t,0) = y(t,1) = 0 & t \in (0,T), \\ y(0,x) = y^0(x) \quad y_t(0,x) = y^1(x) & x \in \Omega, \end{cases}$$

with $h \in L^2(q_T)$ and $(y^0,y^1) \in \mathcal{V}$.
and $R(T;(y^0,y^1))$ defined as follows :

$$R(T;(y^0,y^1)) = \{(y(.,T), y_t(.,T)) : y \textit{ solution of the system with } h \in L^2(q_T)\}.$$

Definitions

Definition 3.1. *System is* ***exactly controllable in time T*** *if and only if : for every initial data* $(y^0,y^1) \in \mathcal{V}$ *we have*

$$R(T;(y^0,y^1)) = \mathcal{V}.$$

Definition 3.2. *System is* ***null controllable in time T*** *if and only if : for every initial data* $(y^0,y^1) \in \mathcal{V}$ *we have*

$$(0,0) \in R(T;(y^0,y^1)).$$

Definition 3.3. *System is* ***approximately controllable in time T*** *if and only if : for every initial data* $(y^0,y^1) \in \mathcal{V}$ *we have*

$$\overline{R(T;(y^0,y^1))} = \mathcal{V}.$$

Definition 3.4. *Assume that* $y^0 \in H_0^1(\Omega)$ *and* $y^1 \in L^2(\Omega)$ *a weak solution on* Q_T *of the boundary-value problem*

$$\begin{cases} y_{tt} - y_{xx} + \xi y = h1_\omega + B & (t,x) \in Q_T, \\ y(t,0) = y(t,1) = 0 & t \in (0,T), \\ y(0,x) = y^0(x) \quad y_t(0,x) = y^1(x) & x \in \Omega, \end{cases}$$

is any function y *satisfying*

$$(y, y_t) \in C(0,T;H_0^1(\Omega)) \cap C(0,T;L^2(\Omega)),$$

and for all $\phi \in \mathcal{C}_c^\infty(Q_T)$

$$\int_{Q_T} (y\phi_{tt} + \nabla y \nabla \phi + \xi y \phi) dxdt = \int_{q_T} h\phi dxdt + \int_{Q_T} B\phi dxdt + \int_\Omega [y\phi_t]_0^T + [y_t\phi]_0^T dxdt$$

we recall the following existence and uniqueness theorem

Theorem 3.1. *Let* $f \in L^2(Q_T)$, $\xi \in L^\infty(Q_T)$ *and* $(y^0,y^1) \in \mathcal{V}$ *then the system*

$$\begin{cases} y_{tt} - y_{xx} + \xi y = f & (t,x) \in Q_T, \\ y(t,0) = y(t,1) = 0 & t \in (0,T), \\ y(0,x) = y^0(x) \quad y_t(0,x) = y^1(x) & x \in \Omega, \end{cases}$$

has a unique weak solutions $y \in C(0,T;H_0^1(\Omega)) \cap C^1(0,T;L^2(\Omega))$ *satisfying the following estimate : there is* $A, B > 0$ *such that for all* $t \in (0,T)$

$$||y, y_t||_{\mathcal{V}} \leq A[||y^0,y^1||_{\mathcal{V}}(1 + ||\xi||_\infty) + ||f||_{L^2(Q_T)}]e^{B\sqrt{||\xi||_\infty}}, \ \ \forall t \in (0,T).$$

From the time-reversibility of the wave equation, we deduce that the notions of Exact controllability and Null controllability are equivalent as states the following proposition :

Proposition 3.1. *The linear wave equation is exactly controllable if and only if it is null controllable.*

Proof: Exact controllability implies Null controllability.
Reciprocally, we suppose that the system is Null controllable : $(0,0) \in R(T;(y^0,y^1))$, from the time-reversibility we can drive the system from $(0,0)$ to any $(y_T^0, y_T^1) \in \mathcal{V}$.
And since :

$$R(T;(y^0,y^1)) = R(T;(0,0)) + S(T)(y^0,y^1).$$

Therefore, the system is exactly controllable. □

Adjoint system

In order to cover the existence of a control for the linear wave equation, we consider the following *Homogeneous system* so called *Adjoint system* :

$$\begin{cases} \phi_{tt} - \Delta\phi + \xi\phi = 0 & (t,x) \in Q_T, \\ \phi(t,0) = \phi(t,1) = 0 & t \in (0,T), \\ \phi(0,x) = \phi^0(x) \quad \phi_t(0,x) = \phi^1(x) & x \in \Omega, \end{cases}$$

where $(\phi^0,\phi^1) \in \mathcal{H} = L^2(\Omega) \times H^{-1}(\Omega)$.
And we define the following functional on $\mathcal{H}$, as follows :

$$\mathcal{J}(\phi^0,\phi^1) = \frac{1}{2}\int_{q_T} |\phi|^2 + \int_{Q_T} B\phi dxdt + <(\phi^0,\phi^1),(y^0,y^1)> .$$

Where ϕ is the solution of the homogeneous sysem with initial data :(ϕ^0,ϕ^1) and $<,>$ is the duality product between $\mathcal{H}$ and $\mathcal{V}$ defined as follows :

$$<(\phi^0,\phi^1),(y^0,y^1)> = \int_\Omega \phi^0 y^1 dx - <\phi^0, y^1>_{H^{-1},H_0^1} .$$

Lemma 3.1. *The control $h \in L^2(q_T)$ drives the initial data (y^0,y^1) to $(0,0)$ if and only if,*

$$\int_{q_T} h\psi dtdx + \int_{Q_T} B\psi dtdx - <(y^0,y^1),(\psi^0,\psi^1)> = 0,$$

for all $(\psi^0,\psi^1) \in \mathcal{H}$, where ψ is the corresponding solution the adjoint system.

Proof: Assume first that $(y^0,y^1),(\phi_T^0,\phi_T^1) \in \mathcal{C}_c^\infty \times \mathcal{C}_c^\infty$, and multiplying the equation of y by ϕ solution to the adjoint system, therefore we integrating by parts, we have

$$\begin{array}{rcl} \int_{q_T} h\phi dtdx + \int_{Q_T} B\phi dtdx & = & \int_{Q_T}[y_{tt} - \Delta y + \xi y]\phi dtdx \\ & = & \int_{Q_T}[\phi_{tt} - \Delta\phi + \xi\phi] y dtdx + \int_\Omega [y_t\phi]_0^T dx - \int_\Omega [y\phi_t]_0^T dx \\ & = & \int_\Omega [\phi_T^0 y_t(T) - \phi_T^1 y(T)]dx - \int_\Omega [\phi(0)y^1 - \phi_t(0)y^0]dx. \end{array}$$

The result extends to arbitrary $(y^0,y^1),(\phi_T^0,\phi_T^1) \in \mathcal{V}$ Hence, we will have

$$\int_{q_T} h\phi dtdx + \int_{Q_T} B\phi dtdx = <(y^0,y^1),(\phi(0),\phi_t(0))> + \int_\Omega \phi_T^0 y_t(T)dx - <\phi_T^1, y(T)>_{H^{-1}(\Omega),H_0^1(\Omega)} .$$

It follows that

$$\int_{q_T} h\psi dtdx + \int_{Q_T} B\psi dtdx - < (y^0, y^1), (\psi^0, \psi^1) >= 0,$$

holds if and only if (y^0, y^1) is controllable to $(0, 0)$ and h is the corresponding control. □

Theorem 3.2. *Let $(y^0, y^1) \in \mathcal{V}$ and suppose that $(\hat{\phi}^0, \hat{\phi}^1) \in \mathcal{H}$ is a minimizer of $\mathcal{J}$. If $\hat{\phi}$ is the corresponding solution of the adjoint system with the initial data $(\hat{\phi}^0, \hat{\phi}^1)$ then,*

$$h = \hat{\phi} 1_\omega.$$

is the minimal L^2-norm control which leads to $(0, 0)$ in time T $\big($if $g \in L^2(q_T)$ any other control which leads the system to $(0, 0)$ then, $||h||_{L^2(q_T)} \leq ||g||_{L^2(q_T)}$ $\big)$.

Proof: Assume that $\mathcal{J}$ has a minimizer $(\hat{\phi}^0, \hat{\phi}^1) \in \mathcal{H}$, then we will have the following Euler-Lagrange equation :

$$\lim_{h \to 0} \frac{1}{h}[\mathcal{J}(\hat{\phi}^0 + h\phi^0, \hat{\phi}^1 + h\phi^1) - \mathcal{J}(\hat{\phi}^0, \hat{\phi}^1)] = 0$$

for all $(\phi^0, \phi^1) \in \mathcal{H}$, where ϕ is the corresponding solution the adjoint system.
Hence,

$$\int_{q_T} \phi\hat{\phi} dtdx + \int_{Q_T} B\phi dtdx - < (y^0, y^1), (\phi^0, \phi^1) >= 0$$

The previous lemma shows that $h = \hat{\phi} 1_\omega$ is a control which leads the data (y^0, y^1) to $(0, 0)$ in time T.
Now, let g a control that leads the system to $(0, 0)$ in time T, we have,

$$\int_{q_T} g\psi dtdx + \int_{Q_T} B\psi dtdx - < (y^0, y^1), (\psi^0, \psi^1) >= 0$$

for all $(\psi^0, \psi^1) \in \mathcal{H}$, where ψ is the corresponding solution the adjoint system.
We take the particular case $(\psi^0, \psi^1) = (\hat{\phi}^0, \hat{\phi}^1)$, and we will obtain,

$$\int_{q_T} |\hat{\phi}|^2 dtdx = \int_{q_T} g\hat{\phi} dtdx$$

Hence, using Cauchy-Schwarz inequality for $h = \hat{\phi} 1_\omega$

$$||h||_{L^2(q_T)} \leq ||g||_{L^2(q_T)}.$$

□

Now, we prove that the control of minimal L^2-norm is determined in a unique way, based on the uniqueness of the mnimum of the functional $\mathcal{J}$. We will prove the uniqueness theorem based the following theorem :

Theorem 3.3. *Let H be a reflexive Banach space, D a closed convex subset of H and $\Gamma : D \to \mathbb{R}$ a function with the following properties :*
1. Γ is strictly convex.
2. Γ is lower semi-continuous.
3. Γ is coercive.
Then Γ attains its unique minimum in D.

Now we can give the following uniqueness theorem of the control of minimal L^2-norm :

Theorem 3.4. *Let $(y^0, y^1) \in \mathcal{V}$ and assume that the observability inequality in time T (.6) holds. Then the functional $\mathcal{J}$ defined as follows $\mathcal{J} : \mathcal{H} \to \mathbb{R}$:*

$$J(\psi^0, \psi^1) = \frac{1}{2} \int_{q_T} |\psi|^2 dtdx + \int_{Q_T} B\psi dtdx - < (y^0, y^1), (\psi^0, \psi^1) > \quad \forall (\psi^0, \psi^1) \in \mathcal{H},$$

has a unique minimizer $(\phi^0, \phi^1) \in \mathcal{H}$.

Proof: Note that $\mathcal{H}$ is a Banach reflexive space, and we take in this case $D = \mathcal{H}$. The functional $\mathcal{J}$ is continuous.
Indeed, let $(\psi^0, \psi^1) \in \mathcal{H}$ and let $(\psi_n^0, \psi_n^1)_{n\in\mathbb{N}} \in \mathcal{H}$, and we assume that :

$$(\psi_n^0, \psi_n^1) \xrightarrow{\mathcal{H}} (\psi^0, \psi^1),$$

the we have,

$$\begin{aligned} |\mathcal{J}(\psi_n^0, \psi_n^1) - \mathcal{J}(\psi^0, \psi^1)| \quad \leq \quad & \tfrac{C_0}{2}||\psi_n - \psi||_{2,q_T} + ||B||_2||\psi_n - \psi||_2 \\ & +||\psi_n^0 - \psi^0||_{H^{-1}(\Omega)}||y^0||_{H_0^1(\Omega)} + ||\psi_n^1 - \psi^1||_{L^2(\Omega)}||y^1||_{L^2(\Omega)}. \end{aligned}$$

Since C_0 is a constant independent of n, when we tend n to $+\infty$ we will deduce that $\mathcal{J}$ is continuous.
In order to prove the strict convexity let $\lambda \in (0,1)$ and $(\psi^0, \psi^1), (\phi^0, \phi^1) \in \mathcal{H}$ we have that,

$$\mathcal{J}(\lambda(\phi^0, \phi^1) + (1-\lambda)(\psi^0, \psi^1)) = \lambda\mathcal{J}(\phi^0, \phi^1) + (1-\lambda)\mathcal{J}(\psi^0, \psi^1) - \frac{\lambda(1-\lambda)}{2}\int_{q_T} |\phi - \psi|^2 dtdx,$$

and due to the continuous dependence we have : there is $C > 0$ such that

$$\int_{q_T} |\phi - \psi|^2 \leq C||(\phi^0 - \psi^0, \phi^1 - \psi^1)||_{\mathcal{H}}^2,$$

then we have for all $\lambda \in (0,1)$:

$$\mathcal{J}(\lambda(\phi^0, \phi^1) + (1-\lambda)(\psi^0, \psi^1)) < \lambda\mathcal{J}(\phi^0, \phi^1) + (1-\lambda)\mathcal{J}(\psi^0, \psi^1),$$

and for the coercivity of $\mathcal{J}$, we have for all $(\psi^0, \psi^1) \in \mathcal{H}$:

$$\mathcal{J}(\psi^0, \psi^1) \geq \frac{1}{2}\int_{q_T} |\psi|^2 - ||B||_2||\psi||_2 - ||(\psi^0, \psi^1)||_{\mathcal{H}}||(y^0, y^1)||_{\mathcal{V}},$$

consequently, $\mathcal{J}$ has a unique minimizer $(\phi^0, \phi^1) \in \mathcal{H}$. □

Observability inequality

In this section we prove the observability inequality, in order to deduce the exact controllability of the linear wave equation. We first prove the estimate of the energy of the adjoint system defined as follows

$$E(t) = \frac{1}{2}[||\phi(t,.)||_{L^2(\Omega)}^2 + ||\phi_t(t,.)||_{H^{-1}(\Omega)}^2],$$

and the energy of the linear wave equation with potential $\xi \in L^\infty(Q_T)$

$$E_0(t) = \frac{1}{2}[||y_t(t,.)||_{L^2(\Omega)}^2 + ||y_x(t,.)||_{L^2(\Omega)}^2].$$

We give these two lemmas in order to prove the *observability inequality* :

Lemma 3.2. *[1]*
Let y be the solution to the following system with $(y^0, y^1) \in \mathcal{V}$

$$\begin{cases} y_{tt} - \Delta y + \xi y = b & (t,x) \in Q_T, \\ y(.,0) = y(.,1) = 0 & t \in (0,T), \\ y(0) = y^0, \ y_t(0) = \phi^1 & x \in \Omega, \end{cases}$$

there exists $C', C'' > 0$ that only depends on Ω and T such that

$$E_0(t) \leq C'(E_0(0)(1 + ||\xi||_\infty) + ||b||_{L^2(Q_T)})e^{C''\sqrt{||\xi||_\infty}}.$$

for every $\xi \in L^\infty(Q_T)$.

Proof: Let $\xi \in L^\infty(Q_T)$ and $\delta = ||\xi||_\infty$. We define the perturbed energy

$$E_\delta(t) = E_0(t) + \frac{\delta}{2}||y(t,.)||^2_{L^2(\Omega)}.$$

we have

$$\begin{array}{rcl} \frac{dE_\delta(t)}{dt} & = & \lim\limits_{h\to 0} \frac{E_\delta(t+h)-E_\delta(t)}{h} \\ & = & \int_\Omega (y_{tt}y_t + y_{x,t}y_x + \delta y_t y)dx \\ & = & \int_\Omega ((\delta-\xi)yy_t + by_t)dx + \Big([y_x y_t]_{\partial\Omega}\Big) \\ & \leq & 2\delta||y(t)||_{L^2(\Omega)}||y_t(t)||_{L^2(\Omega)} + ||b(t)||_{L^2(\Omega)}||y_t(t)||_{L^2(\Omega)} \\ & \leq & (1+2\sqrt{\delta})E_\delta(t) + \frac{1}{2}||b||^2_{L^2(Q_T)} \end{array}$$

then we have

$$E_\delta(t) \leq \big(E_\delta(0) + \frac{1}{2}||b||^2_{L^2(Q_T)}\big)e^{(1+2\sqrt{\delta})t}, \ \ \forall t \in [0,T].$$

hence

$$E_0(t) \leq C'(E_0(0)(1+||\xi||_\infty) + ||b||_{L^2(Q_T)})e^{C''\sqrt{||\xi||_\infty}}, \ \ t \in [0,T].$$

for some $C', C'' > 0$. □

Lemma 3.3. *[1]*
Let ϕ be the solution the following adjoint system

$$\begin{cases} \phi_{tt} - \Delta\phi + \xi\phi = 0 & (t,x) \in Q_T, \\ \phi(.,0) = \phi(.,1) = 0 & t \in (0,T), \\ (\phi(0), \phi_t(0)) = (\phi^0, \phi^1) \in \mathcal{H} & x \in \Omega, \end{cases} \tag{.1}$$

then there exists $C_1, C_2 > 0$ such that for all $t \in [0,T]$:

$$E(t) \leq C_1 E(0)(1+||\xi||^2_\infty)e^{C_2\sqrt{||\xi||_\infty}}.$$

Proof: We define p, q as follows $\phi = p + q$, verifying the following systems :

$$\begin{cases} p_{tt} - \Delta p = 0 & in\, Q_T \\ p(.,0) = p(.,1) = 0 & in\, (0,T) \\ p(0) = \phi^0, \ p_t(0) = \phi^1 & in\, \Omega \end{cases}$$

and

$$\begin{cases} q_{tt} - \Delta q + \xi q = -\xi p & in\, Q_T \\ q(.,0) = q(.,1) = 0 & in\, (0,T) \\ q(0) = 0, \ q_t(0) = 0 & in\, \Omega \end{cases}$$

we multiply the $p_{tt} - \Delta p = 0$ by p_t and we integrate on Ω then we obtain for all $t \in [0,T]$:

$$||p(t,.)||^2_{L^2(\Omega)} + ||p_t(t,.)||^2_{H^{-1}(\Omega)} = 2E(0).$$

we obtain the estimate of the energy of q using Lemma 2.2

$$||q(t,.)||^2_{L^2(\Omega)} + ||q_t(t,.)||^2_{H^{-1}(\Omega)} \leq C||\xi||^2_\infty E(0)e^{C'\sqrt{||\xi||_\infty}}.$$

Hence combining both inequalities we obtain

$$E(t) \leq C_1 E(0)(1+||\xi||^2_\infty)e^{C_2\sqrt{||\xi||_\infty}}.$$

for some $C_1, C_2 > 0$. □

Now we have all the ingredients to prove the observability inequality (.6), and we state the following theorem :

Theorem 3.5. *[1]*
If $T > 2\max(l_1, 1-l_2)$, there exist $A_1, B_1 > 0$ such that

$$||(\phi_0, \phi_1)||^2_{\mathcal{H}} \leq A_1 e^{B_1\sqrt{||\xi||_\infty}} \int_{q_T} |\phi|^2 dxdt,$$

for every $\xi \in L^\infty(Q_T)$ and ϕ solution to the adjoint system with initial conditions $(\phi^0, \phi^1) \in \mathcal{H}$.

Proof: We proceed in several steps :
We define the following sets :

$$\begin{cases} \tau_1(x_0) = \{(x,t) \in (0,x_0) \times (0,T), \;\; x_0 - x < t < T + x - x_0\} \\ \tau_2(x_0) = \{(x,t) \in (x_0,1) \times (0,T), \;\; x - x_0 < t < T + x_0 - x\} \\ \tau(x_0) = \tau_1(x_0) \cup \tau_2(x_0) \end{cases}$$

Step 1 : We prove the following inequality, there is $C > 0$ such that

$$\int_{\tau(x_0)} \phi^2 dxdt \leq Ce^{C\sqrt{||a||_\infty}} \left[\int_0^T |\phi(x_0,t)|^2 dt + ||\phi_x(x_0,.)||^2_{H^{-1}(0,T)} \right].$$

for all $x_0 \in \Omega$ and $T > 2\max(x_0, 1-x_0)$.
We consider ψ solution of the following system

$$\begin{cases} \Delta\psi - \psi_{tt} - \xi\psi = 0 & \textit{in } Q_T \\ \psi(0,.) = \psi(T,.) = 0 & \textit{in } (0,1) \\ \psi(.,x_0) = \phi(.,x_0), \;\; \psi_x(.,x_0) = \phi_x(.,x_0) & \textit{in } (0,T) \end{cases}$$

Now we apply the lemma 2.2 and we obtain

$$E(x) \leq C_1 E(0)(1 + ||\xi||^2_\infty) e^{C_2\sqrt{||\xi||_\infty}}.$$

where $E(x) = \frac{1}{2}[||\phi(.,x)||^2_{L^2(0,T)} + ||\phi_x(.,x)||^2_{H^{-1}(0,T)}]$. Then, if we take $\phi = \psi|_{\tau(x_0)}$, we will get for all $x_0 \in \Omega$:

$$\int_{\tau(x_0)} \phi^2 dxdt \leq Ce^{C\sqrt{||\xi||_\infty}} \left[\int_0^T |\phi(x_0,t)|^2 dt + ||\phi_x(x_0,.)||^2_{H^{-1}(0,T)} \right].$$

Now we integrate with respect to $x_0 \in \omega = (l_1, l_2)$ we obtain, since $\Omega \times (t_1, t_2) \subset \bigcup_{x_0 \in \omega} \tau(x_0)$:

$$\int_{\Omega \times (t_1,t_2)} \phi^2 dxdt \leq Ce^{C\sqrt{||\xi||_\infty}} \left[\int_{q_T} |\phi|^2 dtdx + \int_{l_1}^{l_2} ||\phi_x(.,x)||^2_{H^{-1}(0,T)} dx \right].$$

Since $\Omega \times (t_1, t_2) \subset \bigcup_{x_0 \in \omega} \tau(x_0)$.
Step 2 : We prove

$$\int_{\Omega \times (t_1,t_2)} \phi^2 dxdt \leq Ce^{C\sqrt{||\xi||_\infty}} \int_{q_T} \phi^2 dxdt.$$

In other to obtain an estimate of the term : $\int_\omega ||\phi_x(.,x)||^2_{H^{-1}(0,T)} dx$ we consider the following equation :

$$\begin{cases} \Delta\phi = \phi_{tt} + \xi\phi, \\ \phi \in L^2(q_T). \end{cases}$$

then we will have the following estimate

$$||\Delta\phi||_{L^2(\omega, H^{-2}(0,T))} \leq C(1 + ||\xi||_\infty)||\phi||_{L^2(q_T)}$$

Hence,

$$||\phi_x||_{L^2(\omega, H^{-1}(0,T))} \leq C(1 + \sqrt{||\xi||_\infty})||\phi||_{L^2(q_T)}.$$

we deduce

$$\int_{\Omega\times(t_1,t_2)} \phi^2 dxdt \leq Ce^{C\sqrt{||\xi||_\infty}} \int_{q_T} \phi^2 dxdt.$$

Step 3 : We prove the following inequality

$$\int_{t_1}^{t_2} [||\phi(t,.)||^2_{L^2(\Omega)} + ||\phi_t(t,.)||^2_{H^{-1}(\Omega)}]dt \leq Ce^{C\sqrt{||\xi||_\infty}} \int_{q_T} \phi^2 dxdt.$$

it is sufficient to prove

$$\int_{t_1'}^{t_2'} ||\phi'(t,.)||^2_{H^{-1}(\Omega)} dt \leq C(1+||\xi||_\infty) \int_{\Omega\times(t_1,t_2)} \phi^2 dtdx$$

indeed we multiply the equation $\Delta\phi = \phi_{tt} + \xi\phi$, by $r(t)(-\Delta)^{-1}\phi$ and integrating by parts in $\Omega\times(t_1,t_2)$ we obtain

$$\begin{array}{rcl} \int_{t_1}^{t_2} r(t)||\phi_t(t,.)||_{H^{-1}(\Omega)}dt & = & \int_{t_1}^{t_2} r(t)||\phi(t,.)||^2_{L^2(\Omega)}dt + \int_{\Omega\times(t_1,t_2)} r(t)\xi(x,t)\phi(-\Delta)^{-1}\phi dxdt \\ & & -\int_{t_1}^{t_2} r'(t) < (-\Delta)^{-1}\phi(t,.), \phi_t(t,.) > dt \\ & & -\big[r(t) < (-\Delta)^{-1}\phi(t,.), \phi_t(t,.) > \big]_{t_1}^{t_2}. \end{array}$$

Choosing $r \in C^1([t_1,t_2])$ such that

$$\begin{cases} r(t_1) = r(t_2) = 0 \\ r(t) = 1, \ t \in [t_1', t_2'] \\ \frac{|r'|^2}{r} \in L^\infty(t_1,t_2) \end{cases}$$

where $t_1' = t_1 + \frac{1}{3}(t_2 - t_1)$ and $t_2' = t_2 - \frac{1}{3}(t_2 - t_1)$ We use the fact that $\xi \in L^\infty(Q_T)$ and the boundedness of the operaor $(-\Delta)^{-1}$ we get

$$\begin{array}{rcl} \int_{t_1}^{t_2} r(t)||\phi_t(t,.)||_{H^{-1}(\Omega)}dt & \leq & \int_{t_1}^{t_2} r(t)||\phi(t,.)||_{L^2(\Omega)}dt + ||\xi||_\infty ||(-\Delta)^{-1}|| \int_{(t_1,t_2)} r(t)||\phi||^2_{L^2(\Omega)}dt \\ & & +2||(-\Delta)^{-1}||^2 ||\frac{r'^2}{r}||_\infty \int_{t_1}^{t_2} ||\phi(t,.)||_{L^2(\Omega)}dt + \frac{1}{2}\int_{t_1}^{t_2} r(t)||\phi_t(t,.)||_{H^{-1}(\Omega)}dt \end{array}$$

Hence

$$\int_{t_1'}^{t_2'} ||\phi_t(t,.)||^2_{H^{-1}(\Omega)} dt \leq C(1+||\xi||_\infty) \int_{\Omega\times(t_1,t_2)} \phi^2 dtdx.$$

Then we get

$$\int_{t_1}^{t_2} [||\phi(t,.)||^2_{L^2(\Omega)} + ||\phi_t(t,.)||^2_{H^{-1}(\Omega)}]dt \leq Ce^{C\sqrt{||\xi||_\infty}} \int_{Q_T} \phi^2 dxdt.$$

i.e

$$\int_{t_1}^{t_2} \mathcal{E}(t)dt \leq Ce^{C\sqrt{||\xi||_\infty}} \int_{q_T} \phi^2 dxdt.$$

Based on lemma, and the reversibility of the wave equation we get

$$(t_2 - t_1)\mathcal{E}(0) \leq A(1+||\xi||^2_\infty)e^{B\sqrt{||\xi||_\infty}} \int_{t_1}^{t_2} \mathcal{E}(t)dt.$$

Finally, we get the Observability inequality :

$$||(\phi_0,\phi_1)||^2_{\mathcal{H}} \leq A_1 e^{B_1\sqrt{||\xi||_\infty}} \int_{q_T} |\phi|^2 dxdt.$$

□

4 Estiamte of the minimal L^2-norm control

In this section, we attempt to obtain an estimate of the control of minimal L^2-norm obtained by solving the adjoint system, and the solution of the linearized problem of the semilinear wave equation (.1), where we assume a growth condition (.2) on the nonlinearity f.
Let $\Omega = (0,1)$ and ω an open subset such that $\omega = (l_1, l_2) \subset \Omega$. Let $T > 0$ and $f \in C^1(\mathbb{R})$ verifying $f(0) = 0$, and we define the semilinear wave equation as follows :

$$\begin{cases} y_{tt} - \Delta y + f(y) = h1_\omega + B & (t,x) \in Q_T, \\ y(t,0) = y(t,1) = 0 & t \in (0,T), \\ y(0,x) = y^0(x) \quad y_t(0,x) = y^1(x) & x \in \Omega, \end{cases}$$

where $h \in L^2(q_T)$, $(y^0, y^1) \in \mathcal{V}$ and $B \in L^2(Q_T)$.
In order to linearize th semilinear wave equation, we define the following continuous function g :

$$g(s) = \begin{cases} \frac{f(s)}{s} & s \neq 0, \\ f'(0) & s = 0, \end{cases}$$

which allows us to write the following linearized system : for all $\psi \in L^\infty(Q_T)$:

$$\begin{cases} y_{tt} - \Delta y + g(\psi) y = h1_\omega + B & (t,x) \in Q_T, \\ y(t,0) = y(t,1) = 0 & t \in (0,T), \\ y(0,x) = y^0(x) \quad y_t(0,x) = y^1(x) & x \in \Omega, \end{cases}$$

and now we take the general case of this linear system with $\xi \in L^\infty(Q_T)$:

$$\begin{cases} y_{tt} - \Delta y + \xi y = h1_\omega + B & (t,x) \in Q_T, \\ y(t,0) = y(t,1) = 0 & t \in (0,T), \\ y(0,x) = y^0(x) \quad y_t(0,x) = y^1(x) & x \in \Omega. \end{cases}$$

Proposition 4.1. *Let $\xi \in L^\infty(Q_T)$, $(y^0, y^1) \in \mathcal{V}$, $B \in L^2(Q_T)$ and assume $T > 2\max(l_1, 1 - l_2)$. Then the system is null controllable in time T. and the following estimates holds :*

$$||\phi||_{L^2(q_T)} \leq c_1[||(y^0, y^1)||_{\mathcal{V}} + ||B||_{L^2(Q_T)}(1 + ||\xi||_\infty)^{\frac{1}{2}}]e^{c_2\sqrt{||\xi||_\infty}},$$

for some $c_1, c_2 > 0$, and

$$||(y, y_t)||_{L^\infty(0,T;\mathcal{V})} \leq D_1[||(y^0, y^1)||_{\mathcal{V}}(1 + ||\xi||_\infty)^{\frac{1}{2}} + ||B||_{L^2(Q_T)}(1 + ||\xi||_\infty^2)^{\frac{1}{2}}]e^{D_2\sqrt{||\xi||_\infty}},$$

for some $D_1, D_2 > 0$.

Proof: In order to determine the control of the minimal L^2-norm, we consider the adjoint system as follows :

$$\begin{cases} \phi_{tt} - \Delta\phi + \xi\phi = 0 & (t,x) \in Q_T, \\ \phi(t,0) = 0 = \phi(t,1) = 0 & t \in (0,T), \\ (\phi(0,.), \phi_t(0,.)) = (\phi^0, \phi^1) & x \in \Omega, \end{cases}$$

and the control that we are looking for the the unique solution to this system, where the initial condition (ϕ^0, ϕ^1) is the minimum of the following functional

$$\mathcal{J}(\phi^0, \phi^1) = \frac{1}{2}\int_{q_T} |\phi|^2 dtdx + \int_{Q_T} B\phi dtdx - < (y^0, y^1), (\phi^0, \phi^1) > . \tag{.1}$$

Then, we conclude that ϕ verifies the Euler-Equation, $\forall(\psi^0,\psi^1)\in\mathcal{H}$, where ψ is the solution of the adjoint system with (ψ^0,ψ^1) as initial data

$$\int_{q_T}\phi\psi dtdx+\int_{Q_T}B\psi dtdx-<(y^0,y^1),(\psi^0,\psi^1)>=0.$$

Now we take $(\psi^0,\psi^1)=(\phi^0,\phi^1)$, then we get using Cauchy Schwarz inequality :

$$||\phi||^2_{L^2(q_T)}\leq||B||_{L^2(Q_T)}||\phi||_{L^2(Q_T)}+||(y^0,y^1)||_{\mathcal{V}}||(\phi^0,\phi^1)||_{\mathcal{H}},$$

using lemma 3.3 and we integrate in $(0,T)$:

$$||\phi||^2_{L^2(Q_T)}\leq TC_1||(\phi^0,\phi^1)||^2_{\mathcal{H}}(1+||\xi||^2_\infty)e^{C_2\sqrt{||\xi||_\infty}},$$

Now, and due to the observability inequality (.6), there exists $A_1,B_1>0$ such that :

$$||(\phi^0,\phi^1)||^2_{\mathcal{H}}\leq A_1e^{B_1\sqrt{||\xi||_\infty}}||\phi||^2_{L^2(q_T)},$$

then we conclude that

$$||\phi||_{L^2(q_T)}\leq\sqrt{A_1}[||(y^0,y^1)||_{\mathcal{V}}+\sqrt{C_1T}||B||_{L^2(Q_T)}(1+||\xi||^2_\infty)^{\frac{1}{2}}e^{\frac{C_2}{2}\sqrt{||\xi||_\infty}}]e^{\frac{B_1}{2}\sqrt{||\xi||_\infty}},$$

hence there exists $c_1,c_2>0$ such that

$$||\phi||_{L^2(q_T)}\leq c_1[||(y^0,y^1)||_{\mathcal{V}}+||B||_{L^2(Q_T)}(1+||\xi||^2_\infty)^{\frac{1}{2}}]e^{c_2\sqrt{||\xi||_\infty}}.$$

And using the Lemma 3.2 we obtain, for some $C',C''>0$

$$||(y(t),y'(t))||^2_{\mathcal{V}}\leq C'(||(y^0,y^1)||^2_{\mathcal{V}}(1+||\xi||_\infty)+||\phi1_\omega+B||^2_{L^2(Q_T)})e^{C''\sqrt{||\xi||_\infty}},$$

hence

$$||(y(t),y'(t))||^2_{\mathcal{V}}\leq C'(||(y^0,y^1)||^2_{\mathcal{V}}(1+||\xi||_\infty)+4||B||^2_{L^2(Q_T)}+4||\phi||^2_{q_T})e^{C''\sqrt{||\xi||_\infty}},$$

Therefore, there exist $D_1,D_2>0$ such that

$$||(y,y_t)||_{L^\infty(0,T;\mathcal{V})}\leq D_1[||(y^0,y^1)||_{\mathcal{V}}(1+||\xi||_\infty)^{\frac{1}{2}}+||B||_{L^2(Q_T)}(1+||\xi||^2_\infty)^{\frac{1}{2}}]e^{D_2\sqrt{||\xi||_\infty}}.$$

□

5 The nonlinear operator K

In this section we define the operator K introduced in [1], using the HUM method, that associate to every $\xi \in L^\infty(Q_T)$ the solution y_ξ of the linear wave equation with the potential $g(\xi)$, where K is well defined, continuous and compact, assuming the observability inequality (.6), and due to the Leray-Schaude's degree theorem we obtain the existence of a fixed point for K. We will give under which hypothesis K is a contraction, in order to apply the fixed point Banach theorem, and construct the sequence that converges to the fixed point, in other words to the solution of the semilinear wave equation. We recall that we are dealing with the following system :

$$\begin{cases} y_{tt} - \Delta y + g(\xi)y = h1_\omega & (t,x) \in Q_T, \\ y(t,0) = y(t,1) = 0 & t \in (0,T), \\ y(0,x) = y^0(x) \quad y_t(0,x) = y^1(x) & x \in \Omega, \end{cases}$$

With $\xi \in L^\infty(Q_T)$ and $(y^0, y^1) \in \mathcal{V}$. Our main goal is to find the control $h \in L^2(q_T)$ which drives the system from (y^0, y^1) to zero.
From the previous section, we proved the existence of the control of minimal L^2-norm : $h = \phi$, defined as the solution the homogeneous system, with the initial condition (ϕ^0, ϕ^1) that minimize the functional $\mathcal{J}$ defined previously (.1).

Remark 5.1. *We notice that the the uniqueness of the minimizer of $(\phi^0, \phi^1) \in \mathcal{H}$ of the functional $\mathcal{J}$, and the existence and uniqueness of the solution of the homogeneous system :*

$$\begin{cases} \phi_{tt} - \Delta\phi + g(\xi)\phi = 0 & (t,x) \in Q_T, \\ \phi(t,0) = \phi(t,1) = 0 & t \in (0,T), \\ \phi(0,x) = \phi^0(x) \quad \phi(0,x) = \phi^1(x) & x \in \Omega, \end{cases}$$

And as a result, we constructed the control of minimal L^2-norm, in a unique way, and for every $\xi \in L^\infty(Q_T)$.
Now, we can define the following nonlinear operator :

$$\begin{array}{rccc} K: & L^\infty(Q_T) & \to & C([0,T], H_0^1(\Omega)) \cap C^1([0,T], L^2(\Omega)) \\ & \xi & \mapsto & y_\xi \end{array}$$

where y_ξ is the solution of the following linear wave equation, and $h \in L^2(q_T)$ is the minimal L^2 norm control

$$\begin{cases} y_{\xi,tt} - \Delta y_\xi + g(\xi)y_\xi = h_\xi 1\omega & (t,x) \in Q_T, \\ y_\xi(.,0) = y_\xi(.,1) = 0 & t \in (0,T), \\ (y_\xi(0,.), y_{\xi,t}(0,.)) = (y^0, y^1) & x \in \Omega. \end{cases}$$

Using the embedding $H_0^1(\Omega) \subset L^\infty(\Omega)$, the we can define K as follows :

$$\begin{array}{rccc} K: & L^\infty(Q_T) & \to & L^\infty(Q_T) \\ & \xi & \mapsto & y_\xi \end{array}$$

We have from Proposition 4.1 the following estimate of the $L^\infty(0,T;\mathcal{V})$ of the state y_ξ

$$||(y_\xi, y_{\xi,t})||_{L^\infty(0,T;\mathcal{V})} \leq D_1 ||(y^0, y^1)||_{\mathcal{V}} (1 + ||g(\xi)||_\infty)^{\frac{1}{2}} e^{D_2\sqrt{||g(\xi)||_\infty}},$$

from the growth condition (.2) we deduce from [1] that there exists some $d > 0$ such that

$$||g(\xi)||_\infty \leq 2\beta_0 \log^2(||\xi||_\infty) + d^2, \ \ \forall \xi \in L^\infty(Q_T),$$

hence for some $C > 0$ we will have $e^{D_2\sqrt{||\xi||_\infty}} \leq C(1 + ||\xi||_\infty)^{D_2\sqrt{2\beta_0}}$.
Then we obtain the following estimate

$$||(y_\xi, y_{\xi,t})||_{L^\infty(0,T;\mathcal{V})} \leq CD_1||(y^0, y^1)||_\mathcal{V}(1 + 2\beta_0 \log^2(||\xi||_\infty) + d^2)^{\frac{1}{2}}(1 + ||\xi||_\infty)^{D_2\sqrt{2\beta_0}},$$

as in [1] we take β_0 small enough such that $D_2T\sqrt{2\beta_0} < 1$, we deduce that there exists $M > 0$ verifying

$$CD_1||(y^0, y^1)||_\mathcal{V}(1 + 2\beta_0 \log^2(M) + d^2)^{\frac{1}{2}}(1 + M)^{D_2\sqrt{2\beta_0}} \leq M,$$

hence we get for all $\xi \in L^\infty(Q_T)$ such that $||\xi||_\infty \leq M$ we have $||K(\xi)||_\infty \leq M$.

Proposition 5.1. *Assume the hypothesis of theorem 1.1. Let $M = M(||(y^0, y^1)||_\mathcal{V})$ be the constant defined previously such that $K : B_\infty(0, M) \to B_\infty(0, M)$. Let $\psi^1, \psi^2 \in B_\infty(0, M)$, there exists $c(M) > 0$ such that*

$$||K(\psi^2) - K(\psi^1)||_{L^\infty(Q_T)} \leq c(M)||g'||_\infty||\psi^2 - \psi^1||_{L^\infty(Q_T)}.$$

Proof: In order to prove the continuity of the operator K we consider the perturbed system, through minimal L^2 norm control $h_1, h_2 \in L^2(q_T)$

$$\begin{cases} y_{1,tt} - \Delta y_1 + Ay_1 = h_1 1_\omega & (t,x) \in Q_T, \\ y_1(t,0) = y_1(t,1) = 0 & t \in (0,T), \\ y_1(0,x) = y^0(x) \quad y_{1,t}(0,x) = y^1(x) & x \in \Omega, \end{cases}$$

that we give as follows :

$$\begin{cases} y_{2,tt} - \Delta y_2 + (A+a)y_2 = h_2 1_\omega & (t,x) \in Q_T, \\ y_2(t,0) = y_2(t,1) = 0 & t \in (0,T), \\ y_2(0,x) = y^0(x) \quad y_{2,t}(0,x) = y^1(x) & x \in \Omega, \end{cases}$$

where $A, a \in L^\infty(Q_T)$.
And we define $Z = y_2 - y_1$, verifying the following system :

$$\begin{cases} Z_{tt} - \Delta Z + (A+a)Z = (h_2 - h_1)1_\omega - ay_1 & (t,x) \in Q_T, \\ Z(t,0) = Z(t,1) = 0 & t \in (0,T), \\ Z(0,x) = Z_t(0,x) = 0 & x \in \Omega, \end{cases}$$

the minimal L^2-norm control for both systems are the following :

$$\begin{cases} h_1 = \phi, \\ h_2 = \phi_a, \end{cases}$$

where ϕ and ϕ_a are the solutions to the adjoint systems.
Now, we define $\Phi = \phi_a - \phi$, and then we deduce that Φ solves the following system :

$$\begin{cases} \Phi_{tt} - \Delta\Phi + (A+a)\Phi = -a\phi & (t,x) \in Q_T \\ \Phi(t,0) = \Phi(t,1) = 0 & t \in (0,T) \\ \Phi(0,x) = \phi_a^0 - \phi^0 \quad \Phi_t(0,x) = \phi_a^1 - \phi^1 & x \in \Omega \end{cases}$$

And let $\Phi = \Psi + \overline{\Psi}$ and we get the following systems

$$\begin{cases} \Psi_{tt} - \Delta\Psi + (A+a)\Psi = 0 & (t,x) \in Q_T, \\ \Psi(t,0) = \Psi(t,1) = 0 & t \in (0,T), \\ \Psi(0,x) = \phi_a^0 - \phi^0 \quad \Psi_t(0,x) = \phi_a^1 - \phi^1 & x \in \Omega, \end{cases}$$

and

$$\begin{cases} \overline{\Psi}_{tt} - \Delta\overline{\Psi} + (A+a)\overline{\Psi} = -a\phi & (t,x) \in Q_T, \\ \overline{\Psi}(t,0) = \overline{\Psi}(t,1) = 0 & t \in (0,T), \\ \overline{\Psi}(0,x) = \overline{\Psi}_t(0,x) = 0 & x \in \Omega. \end{cases}$$

Then we deduce that Ψ is the solution to the adjoint system verified by Z : with the potential $A+a$ and the source $\overline{\Psi}1_\omega - ay_1$. We have the following estimates

$$||\phi_a - \phi||_{L^2(q_T)} = ||\Phi||_{L^2(q_T)} \leq ||\Psi||_{L^2(q_T)} + ||\overline{\Psi}||_{L^2(q_T)},$$

and we have from proposition 4.1

$$\begin{aligned} ||\Psi||_{L^2(q_T)} &\leq c_1||\overline{\Psi}1_\omega - ay_1||_{L^2(Q_T)}(1+||A+a||_\infty^2)^{\frac{1}{2}}e^{c_2\sqrt{||A+a||_\infty}} \\ &\leq c_1[||\overline{\Psi}||_{L^2(q_T)} + ||a||_\infty||y_1||_{L^2(Q_T)}](1+||A+a||_\infty^2)^{\frac{1}{2}}e^{c_2\sqrt{||A+a||_\infty}}. \end{aligned}$$

And we have from lemma 3.2

$$||\overline{\Psi}||_{L^2(q_T)} \leq C'||a||_\infty||\phi||_{L^2(Q_T)}e^{C''\sqrt{||A+a||_\infty}},$$

Then we have the estimates of Φ :

$$\begin{aligned} ||\Phi||_{L^2(q_T)} &\leq ||\Psi||_{L^2(q_T)} + ||\overline{\Psi}||_{L^2(q_T)} \\ &\leq C_1(1+||A+a||_\infty^2)^{\frac{1}{2}}[||\overline{\Psi}||_{L^2(q_T)} + ||a||_\infty||y_1||_{L^2(Q_T)}]e^{C_2\sqrt{||A+a||_\infty}} \ , \end{aligned}$$

then we will have

$$||\Phi||_{L^2(q_T)} \leq c_1||a||_\infty(2+||A+a||_\infty^2)^{\frac{1}{2}}\left[||\phi||_{L^2(Q_T)}e^{C''\sqrt{||A+a||_\infty}} + ||y_1||_{L^2(Q_T)}\right]e^{c_2\sqrt{||A+a||_\infty}},$$

and using energy estimate and observability inequality

$$||\phi||_{L^2(Q_T)} \leq \sqrt{C_1A_1}(1+||A+a||_\infty^2)^{\frac{1}{2}}e^{\frac{B_1+C_2}{2}\sqrt{||A+a||_\infty}}||\phi||_{L^2(q_T)}.$$

Then finally we have the estimates of $||y_2 - y_1||_\infty = ||Z||_\infty$, from proposition 4.1

$$||(Z,Z_t)||_{L^\infty(0,T;\mathcal{V})} \leq D_1||a||_\infty[C'||\phi||_{L^2(Q_T)}e^{C''\sqrt{||A+a||_\infty}} + ||y_1||_{L^2(Q_T)}](1+||A+a||_\infty^2)^{\frac{1}{2}}e^{D_2\sqrt{||A+a||_\infty}},$$

hence

$$\begin{aligned} ||(Z,Z_t)||_{L^\infty(0,T;\mathcal{V})} &\leq D_1||a||_\infty[C'\sqrt{C_1A_1}(1+||A+a||_\infty^2)^{\frac{1}{2}}||\phi||_{L^2(q_T)}e^{\frac{B_1+C_2}{2}\sqrt{||A+a||_\infty}}e^{C''\sqrt{||A+a||_\infty}} \\ &\quad +||y_1||_{L^2(Q_T)}](1+||A+a||_\infty^2)^{\frac{1}{2}}e^{D_2\sqrt{||A+a||_\infty}}, \end{aligned}$$

hence for some $D', D'' > 0$

$$||(Z,Z_t)||_{L^\infty(0,T;\mathcal{V})} \leq D'||a||_\infty[(1+||A+a||_\infty^2)^{\frac{1}{2}}||\phi||_{L^2(q_T)} + ||y_1||_{L^2(Q_T)}](1+||A+a||_\infty^2)^{\frac{1}{2}}e^{D''\sqrt{||A+a||_\infty}}.$$

And if we take $\psi^1, \psi^2 \in L^\infty$ and we consider $A = g(\psi^1)$ and $a = g(\psi^2) - g(\psi^1)$ then we will have

$$||(Z,Z_t)||_{L^\infty(0,T;\mathcal{V})} \leq D'||g(\psi^2)-g(\psi^1)||_\infty[(1+||g(\psi^2)||_\infty^2)^{\frac{1}{2}}||\phi||_{L^2(q_T)}+||y_1||_{L^2(Q_T)}](1+||g(\psi^2)||_\infty^2)^{\frac{1}{2}}e^{D''\sqrt{||g(\psi^2)||_\infty}}.$$

Then if we define

$$\zeta(\psi^1,\psi^2) = D'[(1+||g(\psi^2)||_\infty^2)^{\frac{1}{2}}||\phi||_{L^2(q_T)} + ||y_1||_{L^2(Q_T)}](1+||g(\psi^2)||_\infty^2)^{\frac{1}{2}}e^{D''\sqrt{||g(\psi^2)||_\infty}},$$

we will have

$$||K(\psi^2) - K(\psi^1)||_\infty \leq \zeta(\psi^1,\psi^2)||g(\psi^2) - g(\psi^1)||_\infty,$$

hence let $M > 0$ such that there exists $c(M) > 0$ for all $\psi^1, \psi^2 \in B_\infty(0,M)$ we have $\zeta(\psi^1,\psi^2) \leq c(M)$. As a consequence we have

$$||K(\psi^2) - K(\psi^1)||_\infty \leq c(M)||g'||_\infty||\psi^2 - \psi^1||_\infty.$$

□

Remark 5.2. *Assume that $g' \in L^\infty(0, M)$, where $M = M(y^0, y^1)$ as defined earlier, then the operator K that maps $B_\infty(0, M)$ into itself is a contraction under the following hypothesis*

$$\exists\, c(M) > 0 \quad \zeta(\psi^1, \psi^2) \leq c(M), \;\; c(M)||g'||_\infty \leq d < 1.$$

For some $d > 0$, and we will have

$$||K(\psi^2) - K(\psi^1)||_\infty \leq d||\psi^2 - \psi^1||_\infty.$$

Assume the hypothesis of proposition 2.2, we can construct the following sequence that converges to the fixed point of the map K :

$$\begin{cases} \text{For a fixed } \psi_0 \in B_\infty(0, M), \\ \psi_{n+1} = K(\psi_n) \;\; n \geq 0. \end{cases}$$

Consequently, the sequence $(\psi_n)_{n\geq 0}$ converges to the unique fixed point of $K : K(y) = y$. And y satisfies the semilinear wave equation

$$\begin{cases} y_{tt} - \Delta y + f(y) = \phi 1_\omega + B & (t, x) \in Q_T \\ y(t, 0) = y(t, 1) = 0 & t \in (0, T) \\ y(0, x) = y^0(x) \quad y_t(0, x) = y^1(x) & x \in \Omega \end{cases}$$

Where ϕ is the solution of the adjoint system with the initial condition

$$(\phi^0, \phi^1) = \min_{(\psi^0, \psi^1) \in \mathcal{H}} \mathcal{J}(\psi^0, \psi^1)$$

Remark 5.3. *In the general case, the nonlinear operator K, is not a contraction, and as a result we can't converge to the fixed of K and the solution to the semilinear equation wish a particular sequence.*

6 The Newton operator

In this section we will study Newton iteration combined with the continuation method, as part of an iterative procedure to solve the operator equation where $\lambda \in [0,1]$,

$$F(\lambda, u) = 0.$$

We wish to explore the convergence properties of a global Newton iteration (taking place in the in infinite-dimensional function space X), as opposed to fixed point iteration which converges only under special hypothesis. Starting from an adequate choice of the initial guess u^0 we will construct a sequence u^n in X as follows, for all $n \in \mathbb{N}^*$:

$$u^{n+1} = u^n - [D_u F(\lambda, u^n)]^{-1} F(\lambda, u^n),$$

or equivalently

$$[D_u F(\lambda, u^n)](u^{n+1} - u^n) = -F(\lambda, u^n).$$

We are going to prove that this scheme is convergent.

Theorem 6.1. [3]
Assume that $D_U F(\lambda, V)$ is Lipschitz-continuous with respect to V in the ball $S(U, \alpha)$: there is a constant $K > 0$ such that

$$||D_U F(\lambda, V^2) - D_U F(\lambda, V^1)||_{\mathcal{L}(X,\mathcal{X})} \leq K||V^2 - V^1||_X, \ \ \forall V^2, V^1 \in S(U, \alpha).$$

then there exists $0 < \alpha' \leq \alpha$ such that for any initial data $U^0 \in S(U, \alpha')$ the Newton method converges and defines a sequence $(U^n)_{n\in\mathbb{N}} \in S(U, \alpha')$ such that

$$||U^{n+1} - U||_X \leq C||U^n - U||_X^2, \ \ C > 0.$$

If we denote the following spaces :

$$\begin{aligned} X \ = \ & \{(y, \phi) : y \in L^2(Q_T),\ y(.,0) = y(.,1) = 0,\ y_{tt} - \Delta y \in L^2(0,T; H^{-1}(\Omega)),\ (y(0,.), y_t(0,.)) \in \mathcal{V} \\ & \phi \in L^2(q_T)\}, \end{aligned}$$

endowed with the following inner product

$$\begin{aligned} ((y,\phi),(y^*,\phi^*))_X \ = \ & (y, y^*)_2 + \int_0^T < Ly, Ly^* >_{H^{-1}(\Omega), H^{-1}(\Omega)} dt + (\phi, \phi^*)_{L^2(q_T)} \\ & +(y(0,.), y^*(0,.))_{H_0^1(\Omega)} + (y_t(0,.), y_t^*(0,.))_{L^2(\Omega)} \\ & +(y(T,.), y^*(T,.))_{H_0^1(\Omega)} + (y_t(T,.), y_t^*(T,.))_{L^2(\Omega)} \end{aligned}$$

where we define the linear operator $Ly := y_{tt} - \Delta y$, and the notation $(.,.)_2 = (.,.)_{L^2(Q_T)}$ and

$$\mathcal{X} = L^2(0,T; H^{-1}(\Omega)) \times \mathcal{V} \times \mathcal{V}.$$

We can define the Newton operator as follows :

$$\begin{array}{rccl} F: & [0,1] \times X & \mapsto & \mathcal{X} \\ & (\lambda, U) & \to & \big(Ly + \lambda f(y) - \phi 1_\omega, (y(0,.) - y^0, y_t(0,.) - y^1), (y(T,.), y_t(T,.))\big), \end{array}$$

where $U = (y, \phi) \in X$, note that the mapping F is well defined : $\forall(\lambda, U) \in [0,1] \times X$ we have $F(\lambda, U) \in \mathcal{X}$.
We prove that F is a C^1-mapping.
Firstly, F is continuous, indeed, let $(\lambda, U) \in [0,1] \times X$ and $(\lambda_n, U_n) \in [0,1] \times X$ for all $n \geq 0$ such that

$$(\lambda_n, U_n) \xrightarrow{X} (\lambda, U),$$

then we will have

$$\begin{cases} |\lambda_n - \lambda| \longrightarrow 0, \\ ||y_n - y||_{L^2(Q_T)} \longrightarrow 0, \\ ||Ly_n - Ly||_{L^2(0,T;H^{-1}(\Omega))} \longrightarrow 0, \\ ||(y_n(0,.) - y(0,.), y_{n,t}(0,.) - y_t(0,.))||_{\mathcal{V}} \longrightarrow 0, \\ ||(y_n(T,.) - y(T,.), y_{n,t}(T,.) - y_t(T,.))||_{\mathcal{V}} \longrightarrow 0, \\ ||\phi_n - \phi||_{L^2(q_T)} \longrightarrow 0. \end{cases}$$

hence we have,

$$\begin{cases} ||Ly_n + \lambda_n f(y_n) - \phi_n 1_\omega - Ly - \lambda f(y) - \phi 1_\omega||_{L^2(0,T;H^{-1}(\Omega))} \longrightarrow 0, \\ ||(y_n(0,.) - y(0,.), y_{n,t}(0,.) - y_t(0,.))||_{\mathcal{V}} \longrightarrow 0, \\ ||(y_n(T,.) - y(T,.), y_{n,t}(T,.) - y_t(T,.))||_{\mathcal{V}} \longrightarrow 0, \end{cases}$$

we deduce that

$$||F(\lambda_n, U_n) - F(\lambda, U)||_{\mathcal{X}} \longrightarrow 0,$$

hence F is continuous.
Seconldy, we prove that F is differentiable on $[0,1] \times X$.
indeed, let $U = (y, \phi), V = (y^*, \phi^*) \in X$ and $\lambda, \mu \in [0,1]$ then we have

$$\begin{aligned} \lim_{h\to\infty} \frac{F(\lambda+h\mu, U+hV) - F(\lambda,U)}{h} &= \frac{1}{h}[(hLy^* + (\lambda + h\mu)f(y + hy^*) - \lambda f(y) - h\phi^* 1_\omega); \\ & \quad h(y^*|_{t=0}, y_t^*|_{t=0}); h(y^*|_{t=T}, y_t^*|_{t=T}))], \\ &= (Ly^* + \lambda f'(y)y^* + \mu f(y) - \phi^* 1_\omega; (y^*|_{t=0}, y_t^*|_{t=0}); (y^*|_{t=T}, y_t^*|_{t=T})). \end{aligned}$$

hence

$$\begin{array}{rccl} D_{(\lambda,U)}F: & [0,1] \times X & \to & \mathcal{L}([0,1] \times X, \mathcal{X}) \\ & (\lambda, U) & \mapsto & D_{(\lambda,U)}F(\lambda, U) \end{array}$$

where

$$\begin{array}{rccl} D_{(\lambda,U)}F(\lambda,U): & [0,1] \times X & \to & \mathcal{X} \\ & (\mu, y^*, \phi^*) & \mapsto & (Ly^* + \lambda f'(y)y^* + \mu f(y) - \phi^* 1_\omega; (y^*|_{t=0}, y_t^*|_{t=0}); (y^*|_{t=T}, y_t^*|_{t=T})). \end{array}$$

we have $D_{(\lambda,U)}F(\lambda, U) \in \mathcal{L}([0,1] \times X, \mathcal{X})$, indeed for all $U^* = (y^*, \phi^*) \in X$ and $\mu \in [0,1]$

$$\begin{aligned} ||D_{(\lambda,U)}F(\lambda,U)(\mu, U^*)||^2_{\mathcal{X}} &= ||Ly^* + \lambda f'(y)y^* + \mu f(y) - \phi^* 1_\omega||^2_{L^2(0,T;H^{-1}(\Omega))} + ||(y^*|_{t=0}, y_t^*|_{t=0})||^2_{\mathcal{V}} \\ & \quad + ||(y^*|_{t=T}, y_t^*|_{t=T})||^2_{\mathcal{V}} \\ &\leq ||Ly^*||^2_{L^2(0,T;H^{-1}(\Omega))} + ||\phi||^2_{L^2(q_T)} + ||f'||^2_\infty ||y^*||^2_{L^2(Q_T)} + |\mu|^2 ||f||^2_\infty \\ & \quad + ||(y^*|_{t=0}, y_t^*|_{t=0})||^2_{\mathcal{V}} + ||(y^*|_{t=T}, y_t^*|_{t=T})||^2_{\mathcal{V}} \end{aligned}$$

hence

$$||D_{(\lambda,U)}F(\lambda,U)(\mu, U^*)||_{\mathcal{X}} \leq C||(\mu, U^*)||,$$

where $C = \max(1, ||f||_\infty, ||f'||_\infty)$ and $||(\mu, U^*)|| = \sqrt{|\mu|^2 + ||U^*||^2_X}$.
we deduce that $D_{(\lambda,U)}F$ is well defined.
to conclude that F is a C^1 mapping, we need to prove that $D_{(\lambda,U)F}$ is a lipschitz mapping (in order to use theorem [3]).
Let $(\lambda^1, U^1), (\lambda^0, U^0), (\mu, U^*) \in [0,1] \times X$ then we have

$$\begin{aligned} ||(D_{(\lambda,U)}F(\lambda^1, U^1) - D_{(\lambda,U)}F(\lambda^0, U^0))(\mu, U^*)||_{\mathcal{X}} &= ||(\lambda^1 f'(y^1) - \lambda^0 f'(y^0))y^* \\ & \quad + \mu(f(y^1) - f(y^0))||_{L^2(0,T;H^{-1}(\Omega))} \\ &\leq ||(f'(y^1) - f'(y^0))y^*||_{L^2(Q_T)} + ||f'||_\infty |\lambda^1 - \lambda^0| \\ & \quad + ||f(y^1) - f(y^0)||_{L^2(Q_T)}. \end{aligned}$$

now if we assume the following hypothesis

$$\exists K > 0,\ |f(t) - f(s)| \leq K|t-s|\ \ \forall t, s \in \mathbb{R}. \quad (\mathcal{H}_1)$$

then we will have

$$\begin{array}{rcl} ||D_{(\lambda,U)}F(\lambda^1, U^1) - D_{(\lambda,U)}F(\lambda^0, U^0)||_{\mathcal{L}([0,1]\times X,\mathcal{X}} & \leq & K||y^1 - y^0||_{L^2(Q_T)} + ||f'||_\infty|\lambda^1 - \lambda^0| \\ & & +||f'(y^1) - f'(y^0)||_\infty \end{array}$$

and assuming

$$\exists K' > 0,\ |f'(t) - f'(s)| \leq K'|t-s|\ \ \forall t, s \in \mathbb{R}. \quad (\mathcal{H}_2)$$

then we deduce that

$$||D_{(\lambda,U)}F(\lambda^1, U^1) - D_{(\lambda,U)}F(\lambda^0, U^0)||_{\mathcal{L}([0,1]\times X,\mathcal{X}} \leq 2\max(K, C)||y^1 - y^0||_{L^2(Q_T)} + ||f'||_\infty|\lambda^1 - \lambda^0|$$

hence F is a C^1 mapping on $[0,1] \times X$.
Now for a fixed $\lambda \in [0,1]$ let $U(\lambda) = (\lambda, y(\lambda), \phi(\lambda))$ the solution to $F(\lambda, U(\lambda)) = 0$.
we have $[D_U F(\lambda, U(\lambda))]^{-1} \in \mathcal{L}(\mathcal{X}, X)$, indeed we have

$$\begin{array}{rccl} D_U F(\lambda, U): & X & \to & \mathcal{X} \\ & (y^*, \phi^*) & \mapsto & (Ly^* + \lambda f'(y)y^* - \phi^* 1_\omega; (y^*|_{t=0}, y_t^*|_{t=0}); (y^*|_{t=T}, y_t^*|_{t=T})). \end{array}$$

We first prove that $D_U F(\lambda, U)$ is one-to-one, let $(z, \psi) \in X$ such that

$$D_U F(\lambda, U)(z, \psi) = 0_{\mathcal{X}},$$

which is equivalent to,

$$\begin{cases} Lz + \lambda f'(y)z = \psi 1_\omega & in\ \ L^2(0,T; H^{-1}(\Omega)), \\ z|_{\Sigma_T} = 0 & t \in (0,T), \\ (z(0,.), z_t(0,.)) = (0,0) & x \in \Omega, \\ (z(T,.), z_t(T,.)) = (0,0) & x \in \Omega. \end{cases}$$

since $f'(y) \in L^\infty(Q_T)$ and $\phi \in L^2(q_T)$, we deduce that there is a unique weak solution $z \in C([0,T], H^{-1}(\Omega)) \cap C^1([0,T], L^2(\Omega))$, and also 0_X is a solution to the system, hence by the argument of uniqueness $D_U F(\lambda, U)$ is one-to-one.
Secondly, we prove that $D_U F(\lambda, U)$ is onto : Let $(A, B, C) \in \mathcal{X}$ (where $B = (B^0, B^1)$ and $C = (C^0, C^1)$), is there $(z, \psi) \in X$ such that

$$(\Gamma): \begin{cases} Lz + \lambda f'(y)z = \psi 1_\omega + A & in\ \ L^2(Q_T), \\ z(.,0) = z(.,1) = 0 & t \in (0,T), \\ (z(0,.), z_t(0,.)) = (B^0, B^1) & x \in \Omega, \\ (z(T,.), z_t(T,.)) = (C^0, C^1) & x \in \Omega. \end{cases}$$

let $\psi \in L^2(q_T)$ the conrol of the minimal L^2-norm, satisfying the following system

$$\begin{cases} L\phi + \lambda f'(y)\phi = 0 & in\ \ L^2(Q_T), \\ \phi(.,0) = \phi(.,1) = 0 & t \in (0,T), \\ (\phi(0,.), \phi_t(0,.)) = (\phi^0, \phi^1) & x \in \Omega, \end{cases}$$

where

$$(\phi^0, \phi^1) = \text{argmin}_{(\psi^0,\psi^1)\in\mathcal{H}} J(\psi^0, \psi^1) = \frac{1}{2}\int_{q_T} |\psi|^2 dxdt - \int_{Q_T} A\psi dxdt + < (y^0, y^1), (\psi^0, \psi^1) >$$

where ψ is the solution to the adjoint system, with (ψ^0, ψ^1) as initial conditions.
hence the system (Γ) has a unique solution $z \in C([0,T], H^{-1}(\Omega)) \cap C^1([0,T], L^2(\Omega))$, we deduce the

exitence of $(z,\psi)\in X$ such that $D_UF(\lambda,U)(z,\psi)=(A,B,C)$.
we conclude that $D_UF(\lambda,U)$ is one-to-one and onto, hence $[D_UF(\lambda,U(\lambda))]^{-1}\in\mathcal{L}(\mathcal{X},X)$.
Now we can define Newton's method : For an adequate choice of U^0 we have for all $n\geq 1$

$$U^{n+1}=U^n-[D_UF(\lambda,U^n)]^{-1}F(\lambda,U^n),$$

which is equivalent to

$$D_UF(\lambda,U^n)(U^{n+1}-U^n)=-F(\lambda,U^n).$$

Now let $U^0=(y^0,\phi^0)\in X$ the solution to the following system

$$\begin{cases} Ly^0=\phi^0 1_\omega & in\ L^2(Q_T),\\ y^0(.,0)=y^0(.,1)=0 & in\ (0,T),\\ (y^0(0,.),y^0_t(0,.))=(y^0,y^1) & in\ \Omega.\end{cases}$$

where ϕ^0 is the control of the minimal L^2-norm.
the Newton algorithm is the following : for all $n\geq 0$,

$$\begin{cases} Ly^{n+1}+\lambda f'(y^n)u^{n+1}=\phi^{n+1}1_\omega+\lambda(f'(y^n)y^n-f(y^n)) & in\ L^2(Q_T),\\ y^{n+1}(.,0)=y^{n+1}(.,1)=0 & in\ (0,T),\\ (y^{n+1}(0,.),y^{n+1}_t(0,.))=(y^0,y^1) & in\ \Omega.\\ (y^{n+1}(T,.),y^{n+1}_t(T,.))=(0,0) & in\ \Omega.\end{cases}$$

where ϕ^{n+1} is the control of minimal L^2-norm, that drives the system from (y^0,y^1) to zero.
we define the following notations : for all $n\geq 1$,

$$\begin{cases} a=\lambda[f'(y^n)-f'(y^{n-1})],\\ b=\lambda[f'(y^n)-f'(y^{n-1})+f(y^{n-1})-f(y^n)].\end{cases}$$

then we will have the following estimates :

$$\begin{aligned} ||\phi^{n+1}-\phi^n||_{L^2(q_T)} &\leq C_1(2+\lambda||f'(u^n)||^2_\infty)e^{C_2\sqrt{\lambda||f'(f^n)||^2_\infty}}\\ &\quad\times\Bigg[||b||_{L^2(Q_T)}+||a||_\infty\Bigg(||u^n||_{L^2(Q_T)}+C_1(1+\lambda||f'(u^{n-1})||_\infty)^{\frac{1}{2}}\\ &\quad\times e^{C_2\sqrt{\lambda||f'(u^{n-1})||_\infty}}.e^{C_2\sqrt{\lambda||f'(u^n)||_\infty}}\Bigg)||\phi^n||_{L^2(q_T)}\Bigg].\end{aligned}$$

and

$$\begin{aligned} ||(y^{n+1}-y^n),(y^{n+1}-y^n)_t||_\infty &\leq C_1.\lambda||f(y^{n-1})-f(y^n))+f'(y^{n-1})(y^n-y^{n-1})||_{L^2(Q_T)}.e^{C_2\sqrt{\lambda||f'(y^n)||_\infty}}\\ &\leq C_1.T.\lambda\Big[||f(y^{n-1})-f(y^n))||_\infty+||f'(y^{n-1})(y^n-y^{n-1})||_\infty\Big]\\ &\quad\times e^{C_2\sqrt{\lambda||f'(y^n)||_\infty}}.\end{aligned}$$

Let $n\geq 1$, we have

$$\begin{aligned} U^{n+1}-U &= U^n-[D_UF(\lambda,U^n)]^{-1}(F(\lambda,U^n)-F(\lambda,U))-U,\\ &= [D_UF(\lambda,U^n)]^{-1}[F(\lambda,U)-F(\lambda,U^n)-D_UF(\lambda,U^n)(U^n-U)],\\ &= [D_UF(\lambda,U^n)]^{-1}\int_0^1[D_UF(\lambda,U^n+t(U^n-U))-D_UF(\lambda,U^n)](U^n-U)dt.\end{aligned}$$

hence

$$\begin{aligned} ||U^{n+1}-U||_X &\leq ||[D_UF(\lambda,U^n)]^{-1}||_{\mathcal{L}(\mathcal{X},X)}\int_0^1 Kt||U^n-U||^2_X dt,\\ &\leq \tfrac{K}{2}||[D_UF(\lambda,U^n)]^{-1}||_{\mathcal{L}(\mathcal{X},X)}||U^n-U||^2_X.\end{aligned}$$

6.1 Mixed formulation of the controllability problem

In this section we introduce a different approach, so called *mixed formulation*, that will help us interpret the theoretical results of the last sections numerically. As described at length in previous sections, that the controllability problem of the wave equation, and the minimal L^2 norm control is equivalent to the following problem

$$(\mathcal{P}) : \min_{(\phi^0,\phi^1)\in\mathcal{H}} J^*(\phi^0,\phi^1) = \frac{1}{2}\int_{q_T} |\phi|^2 dxdt + < \phi^1, y^0 >_{H^{-1}(\Omega),H_0^1(\Omega)} - \int_0^1 \phi^0 y^1 dx,$$

We seek minimizers of $\mathcal{J}^*$ over a class of admissible functions : ϕ the solution of the following system

$$\begin{cases} L\phi(t,x) = 0 & (t,x) \in (0,T)\times\Omega, \\ \phi(t,0) = \phi(t,1) = 0 & t \in (0,T), \\ \phi(0,x) = \phi^0(x),\ \phi_t(0,x) = \phi^1(x) & x \in (0,1). \end{cases}$$

we define $L\phi := \phi_{tt} - \Delta\phi + \xi\phi$ where $\xi \in L^\infty(Q_T)$, now if we consider the following problem

$$(\mathcal{P}') : \min_{\phi\in W} \hat{J}^*(\phi) = \frac{1}{2}\int_{q_T} |\phi|^2 dxdt + < \phi_t(0,.), y^0) > - \int_0^1 \phi(0,.) y^1 dx,$$

where

$$W = \{\phi : \phi \in L^2(q_T),\ \phi(.,0) = \phi(.,1) = 0,\ L\phi = 0 \in L^2(0,T;H^{-1}(\Omega))\},$$

note that W is a subspace of the following Hilbert space

$$\Phi = \{\phi : \phi \in L^2(q_T),\ \phi(.,0) = \phi(.,1) = 0,\ L\phi \in L^2(0,T;H^{-1}(\Omega))\},$$

equipped with the scalar product : for all $(\phi,\psi) \in \Phi$ and $\mu > 0$:

$$(\phi,\psi)_\Phi = \int_{q_T} \phi\psi dxdt + \mu\int_0^T < L\phi, L\psi >_{H^{-1}(\Omega),H^{-1}(\Omega)} dt.$$

we deduce that the problems $\mathcal{P}$ and $\mathcal{P}'$ are equivalent.

Remark 6.1. *Assume that $\phi \in W$ then we have $(\phi(0,.),\phi_t(0,.)) \in \mathcal{H}$, which follows from the following generalized inequality observability*[5] *: $\exists C > 0$ such that*

$$||(\phi(0,.),\phi_t(0,.))||^2_{\mathcal{H}} \le C\Big(||\phi||^2_{L^2(q_T)} + ||L\phi||^2_{L^2(0,T;H^{-1}(\Omega))}\Big),\ \ \forall\phi \in \Phi.$$

Since the main variable now is ϕ, and with the constraint $L\phi = 0 \in L^2(0,T;H^{-1}(\Omega))$ we introduce the Lagrange multiplier $\lambda \in L^2(0,T;H_0^1(\Omega))$ in the following way, where we consider the following problem

$$\begin{cases} a(\phi,\psi) + b(\psi,\lambda) &= l(\psi) \quad \forall\psi \in \Phi, \\ b(\phi,\eta) &= 0 \quad \forall\eta \in L^2(0,T;H_0^1(\Omega)). \end{cases}$$

where

$$\begin{array}{rcl} a : \Phi\times\Phi & \to & \mathbb{R} \\ (\phi,\psi) & \mapsto & \int_{q_T} \phi\psi dxdt. \end{array}$$

and

$$\begin{array}{rcl} b : \Phi\times L^2(0,T;H_0^1(\Omega)) & \to & \mathbb{R} \\ (\phi,\lambda) & \mapsto & \int_0^T < L\phi,\lambda >_{H^{-1}(\Omega),H_0^1(\Omega)} dt. \end{array}$$

and

$$\begin{array}{rcl} l : \Phi & \to & \mathbb{R} \\ \phi & \mapsto & \int_0^1 \phi(0,.) y^1 dx - < \phi_t(0,.), y^0 >_{H^{-1}(\Omega),H_0^1(\Omega)}. \end{array}$$

we have the following result :

Theorem 6.2. [6]

— *The mixed formulation is well-posed.*
— *The unique solution* $(\phi, \lambda) \in \Phi \times L^2(0, T; H_0^1(\Omega))$ *is the unique saddle point of the Lagrangian* $\mathcal{L} : \Phi \times L^2(0, T; H_0^1(\Omega)) \to \mathbb{R}$ *defined by*

$$\mathcal{L}(\phi, \lambda) = \frac{1}{2} a(\phi, \phi) + b(\phi, \lambda) + l(\phi).$$

— *The optimal function* ϕ *is the minimizer of* $\hat{J}^*$ *over* Φ *while the optimal function* $\lambda \in L^2(0, T; H_0^1(\Omega))$ *is the state of the controlled wave equation in the weak sense.*

6.2 Numerical results

We illustrate some of these concepts in the context of a model semilinear hyperbolic partial differential equation in one dimension. Our numerical solution is via a conventional finite-element method.
we present an approach in order to use finite volume method in FreeFem++. To this end, we must restrict our study to a rectangular domain $ABCD$ $(A(0,0), B(0,2), C(1,2), D(1,0))$ with a structured mesh (cf. Figure 5.1). We have chosen to implement a conforming finite-element method, with rectangular elements, we adopt a triangular *HCT element* (Hsieh-Clough-Tocher C^1 finite element) : Let us call $\mathcal{T}_h^*$ the sub mesh of $\mathcal{T}_h$ where all triangles are split in 3 at the barycenter

$$PHCT_h = \{v \in C^1(\Omega), \forall K \in \mathcal{T}_h^*, v|_K \in P_3\},$$

where P_3 is the set of polynomials of $\mathbb{R}^2$ of degrees ≤ 3.
we chose the following mesh of the doamin $(0,T) \times (0,1)$ where $T = 2$, and $\omega = (0.2, 0.4)$, with the subdivision $(\delta x, \delta t) = (\frac{1}{60}, \frac{2}{120})$, where we obtain 16098 triangles.

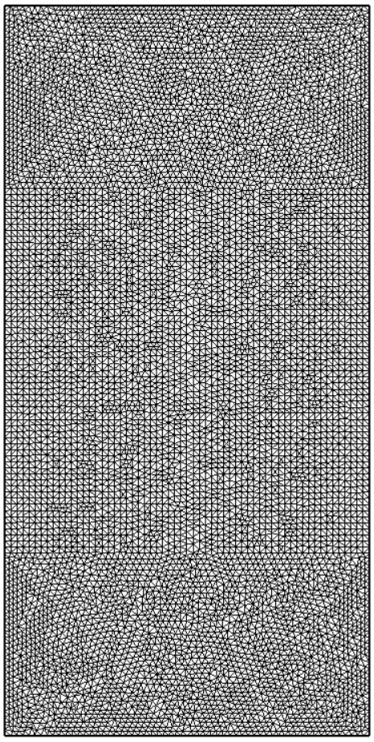

FIGURE 6.1 – $ABCD$ with the associated mesh $\mathcal{T}_h$.

```
/// THE TRIANGULATION ///

border C0(s=0, 1){x=s;  y=0;      label=Nbas;}
border C1(s=0, 2){x=1;  y=s;      label=Ndroit;}
border C2(s=0, 1){x=1-s; y=2;  label=Nhaut;}
border C3(s=0, 2){x=0.; y=2-s;  label=Ngauche;}

int n=60, m1=120;

mesh Th=buildmesh(C0(n) + C1(m1) + C2(n) + C3(m1));
```

now, we define the initial condition to the Newton algorithm, which is the solution to the following system

$$\begin{cases} u^0_{tt} - \Delta u^0 = \phi^0 1_\omega, & (t,x) \in (0,T) \times (0,1) \\ u^0(.,0) = u^0(.,1) = 0, & t \in (0,1) \\ (u^0(.,0), u^0_t(.,0)) = (y^0, y^1), & x \in (0,1) \end{cases}$$

with the initial condition $(y^0(x), y^1(x)) = (\sin(\pi x), 0)$ (cf. Figure 5.2).

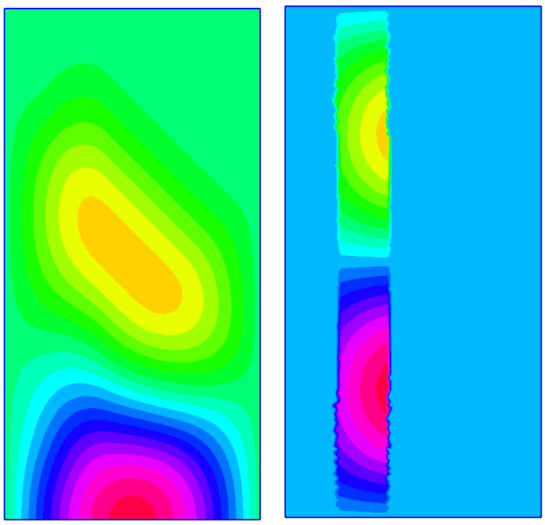

FIGURE 6.2 – the solution (state) u^0, and the restriction of the minimal L^2 norm control on ω.

```
/// Initialization problem ///
problem mixedwaveweakL([phi,phix,phiy,lambdaY],
                       [phit,phixt,phiyt,lambdaYt])=

   int2d(Th,qft=qfHCT5)(omega*phi*phit)
 + int2d(Th,qft=qfHCT5)(ropt* (dyy(phi)-dxx(phi))*(dyy(phit)-dxx(phit)))

 - int2d(Th,qft=qfHCT5) (lambdaY*dyy(phit)-lambdaY*dxx(phit))
 - int2d(Th,qft=qfHCT5) (lambdaYt*dyy(phi)-lambdaYt*dxx(phi))

 + int1d(Th,qft=qfHCT5,Ngauche)(pena*(phi*phit+phiy*phiyt))
 + int1d(Th,qft=qfHCT5,Ndroit)(pena*(phi*phit+phiy*phiyt))

// + int1d(Th,qft=qfHCT5,Nhaut)(lambdaY*dy(phit))
 - int1d(Th,qft=qfHCT5,Nbas)(y0*dy(phit))

 + on(Ngauche,lambdaY=0) + on(Ndroit,lambdaY=0)
 + on(Nbas,lambdaY=y0)+ on(Nhaut,lambdaY=0);


mixedwaveweakL;
```

Now we take a special case, a smooth bounded nonlinearity $f \in C^\infty(\mathbb{R})$ where the method of continuation is not needed, and we keep the same initial conditions : $(y^0(x), y^1(x)) = (\sin(\pi x), 0)$, we take $f(s) = 300\sin(s)$, then we will have the following results

iteration k	$\frac{\|\|y_k - y_{k-1}\|\|_{L^2(Q_T)}}{\|\|y_k\|\|_{L^2(Q_T)}}$	$\frac{\|\|\phi_k - \phi_{k-1}\|\|_{L^2(q_T)}}{\|\|\phi\|\|_{L^2(q_T)}}$	$\|\|y_k\|\|_{L^2(Q_T)}$
1	2.26361	15.3425	0.891279
2	0.571101	0.689102	0.921963
3	0.18021	0.626407	0.830907
4	0.0802735	0.142607	0.794898
5	0.0346593	0.0910415	0.779976
6	0.0181744	0.0481547	0.772775
7	0.00929846	0.023272	0.769341
8	0.00473192	0.0113813	0.767693
9	0.00239554	0.00562427	0.766891
10	0.00120894	0.00279933	0.766495
11	0.000610372	0.00140209	0.766298
12	0.000309347	0.000707057	0.766198
13	0.000157794	0.00035931	0.766147
14	8.11497e-05	0.000184145	0.766122

TABLE 6.1 – Norms of the difference between y_k, y_{k-1} the state and ϕ_k, ϕ_{k-1} the control defined by Newton algorithm

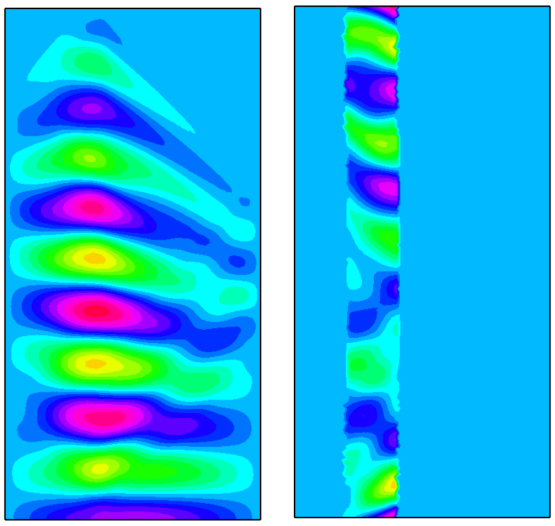

FIGURE 6.3 – The solution (state) and the restriction of the minimal L^2 norm control on ω. in the case : $f(s) = 300\sin(s)$

we define the nonlinearity as follows in the *Freefem*++ code

```
real alpha=300.;

cout << "alpha=" << alpha << endl;
macro g(s)    alpha*sin(s)   //
macro gp(s) alpha*cos(s) //
```

and we modify the previous mixed formulation in order to solve the nonlinear problem based on the Newton method.

```
problem mixedwaveweak([phi,phix,phiy,lambdaY],
               [phit,phixt,phiyt,lambdaYt],solver="UMFPACK")=

    int2d(Th,qft=qfHCT5)(omega*phi*phit)
  + int2d(Th,qft=qfHCT5)(ropt* (dyy(phi)-dxx(phi)
  +ss*gp(lambdak)*phi)*(dyy(phit)-dxx(phit)+ss*gp(lambdak)*phit))

  - int2d(Th,qft=qfHCT5) (lambdaY*dyy(phit)+dx(lambdaY)*dx(phit)
  +lambdaY*ss*gp(lambdak)*phit)
  - int2d(Th,qft=qfHCT5) (lambdaYt*dyy(phi)+dx(lambdaYt)*dx(phi)
  +lambdaYt*ss*gp(lambdak)*phi)

  + int1d(Th,qft=qfHCT5,Ngauche)(pena*(phi*phit+phiy*phiyt))
  + int1d(Th,qft=qfHCT5,Ndroit)(pena*(phi*phit+phiy*phiyt))

  + int2d(Th,qft=qfHCT5)((-ss*g(lambdak)+ss*gp(lambdak)*lambdak)*phit)

 // + int1d(Th,qft=qfHCT5,Nhaut)(lambdaY*dy(phit))
   - int1d(Th,qft=qfHCT5,Nbas)(y0*dy(phit))

  + on(Ngauche,lambdaY=0) + on(Ndroit,lambdaY=0)
  + on(Nbas,lambdaY=y0)+ on(Nhaut,lambdaY=0);




cout << "valeur de ss=" << ss << endl;

err=1; nn=0;


while (nn< niter && err > 1e-4)

     {

 cout << "valeur de nn=" << nn << endl;

mixedwaveweak;
```

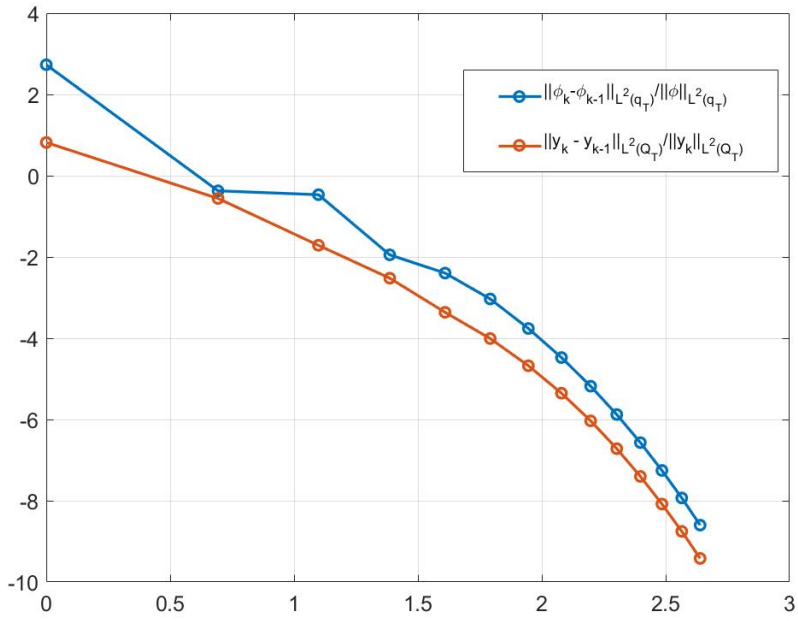

FIGURE 6.4 – The convergence towards zero of $\frac{||y_k - y_{k-1}||L^2(Q_T)}{||y_k||_{L^2(Q_T)}}$.

Now we introduce a case where the method of continuation is necessary in order for the Newton method converges in less iterations : we take the following nonlinearity $f(s) = 300s\log^2(1+|s|)$, with the same initial conditions $(y^0(x), y^1(x)) = (\sin(x), 0)$.

iteration k	$\frac{\|\|y_k - y_{k-1}\|\|_{L^2(Q_T)}}{\|\|y_k\|\|_{L^2(Q_T)}}$	$\frac{\|\|\phi_k - \phi_{k-1}\|\|_{L^2(q_T)}}{\|\|\phi\|\|_{L^2(q_T)}}$	$\|\|y_k\|\|_{L^2(Q_T)}$
1	0.967778	2.41487	0.45659
2	1.53725	3.41094	0.816187
3	0.642466	1.65259	0.731839
4	0.736727	1.03219	0.759389
5	0.377198	0.759663	0.659835
6	0.20499	0.692112	0.604856
7	0.152716	0.309567	0.587966
8	0.0699938	0.164641	0.571493
9	0.0529491	0.107872	0.563602
10	0.0286655	0.0703918	0.557775
11	0.0250302	0.0572771	0.553777
12	0.0234704	0.0532546	0.550012
13	0.0265217	0.0562333	0.546219
14	0.0337966	0.0669183	0.541757
15	0.0495198	0.0895312	0.536097

iteration k	$\frac{\|\|y_k - y_{k-1}\|\|_{L^2(Q_T)}}{\|\|y_k\|\|_{L^2(Q_T)}}$	$\frac{\|\|\phi_k - \phi_{k-1}\|\|_{L^2(q_T)}}{\|\|\phi\|\|_{L^2(q_T)}}$	$\|\|y_k\|\|_{L^2(Q_T)}$
16	0.0798702	0.130524	0.528958
17	0.114991	0.176452	0.521938
18	0.0980822	0.159105	0.515323
19	0.0911878	0.15763	0.510118
20	0.138086	0.226905	0.507052
21	0.208044	0.39744	0.503291
22	0.0858205	0.269014	0.491075
23	0.128236	0.266391	0.498805
24	0.127562	0.291216	0.491369
25	0.136801	0.308996	0.499291
26	0.138862	0.317166	0.491564
27	0.141452	0.32482	0.499734
28	0.14076	0.321927	0.49162
29	0.143398	0.32937	0.499929
30	0.141918	0.324816	0.491664
31	0.144429	0.332012	0.500034
32	0.142501	0.326263	0.491688
33	0.144988	0.333406	0.500091
34	0.142821	0.327054	0.491701
35	0.145293	0.334169	0.500122
36	0.142993	0.32748	0.491709
37	0.145458	0.334582	0.500139
38	0.143086	0.327711	0.491713
39	0.145549	0.334807	0.500148
40	0.143137	0.327837	0.491715
41	0.145598	0.334929	0.500153
42	0.143165	0.327905	0.491716
43	0.145624	0.334995	0.500156
44	0.14318	0.327942	0.491717
45	0.145639	0.335031	0.500157
46	0.143188	0.327962	0.491717
47	0.145647	0.335051	0.500158
48	0.143193	0.327973	0.491717
49	0.145651	0.335062	0.500158
50	0.143195	0.327979	0.491717
51	0.145653	0.335068	0.500158
52	0.143196	0.327982	0.491717
53	0.145655	0.335071	0.500159
54	0.143197	0.327984	0.491718
55	0.145655	0.335072	0.500159
56	0.143197	0.327985	0.491718
57	0.145656	0.335073	0.500159
58	0.143198	0.327986	0.491718
59	0.145656	0.335074	0.500159
60	0.143198	0.327986	0.491718

iteration k	$\frac{\|\|y_k - y_{k-1}\|\|_{L^2(Q_T)}}{\|\|y_k\|\|_{L^2(Q_T)}}$	$\frac{\|\|\phi_k - \phi_{k-1}\|\|_{L^2(q_T)}}{\|\|\phi\|\|_{L^2(q_T)}}$	$\|\|y_k\|\|_{L^2(Q_T)}$
61	0.145656	0.335074	0.500159
62	0.143198	0.327986	0.491718
63	0.145656	0.335074	0.500159
64	0.143198	0.327986	0.491718
65	0.145656	0.335074	0.500159
66	0.143198	0.327986	0.491718
67	0.145656	0.335074	0.500159
68	0.143198	0.327986	0.491718
69	0.145656	0.335074	0.500159
70	0.143198	0.327986	0.491718
71	0.145656	0.335074	0.500159
72	0.143198	0.327986	0.491718
73	0.145656	0.335074	0.500159
74	0.143198	0.327986	0.491718
75	0.145656	0.335074	0.500159
76	0.143198	0.327986	0.491718
77	0.145656	0.335074	0.500159
78	0.143198	0.327986	0.491718
79	0.145656	0.335074	0.500159
80	0.143198	0.327986	0.491718
81	0.145656	0.335074	0.500159
82	0.143198	0.327986	0.491718
83	0.145656	0.335074	0.500159
84	0.143198	0.327986	0.491718
85	0.145656	0.335074	0.500159
86	0.143198	0.327986	0.491718
87	0.145656	0.335074	0.500159
88	0.143198	0.327986	0.491718
89	0.145656	0.335074	0.500159
90	0.143198	0.327986	0.491718
91	0.145656	0.335074	0.500159
92	0.143198	0.327986	0.491718
93	0.145656	0.335074	0.500159
94	0.143198	0.327986	0.491718
95	0.145656	0.335074	0.500159
96	0.143198	0.327986	0.491718
97	0.145656	0.335074	0.500159
98	0.143198	0.327986	0.491718
99	0.145656	0.335074	0.500159
100	0.143198	0.327986	0.491718
101	0.145656	0.335074	0.500159
102	0.143198	0.327986	0.491718
103	0.145656	0.335074	0.500159
104	0.143198	0.327986	0.491718
105	0.145656	0.335074	0.500159
106	0.143198	0.327986	0.491718
107	0.145656	0.335074	0.500159
108	0.143198	0.327986	0.491718

iteration k	$\frac{\|\|y_k - y_{k-1}\|\|_{L^2(Q_T)}}{\|\|y_k\|\|_{L^2(Q_T)}}$	$\frac{\|\|\phi_k - \phi_{k-1}\|\|_{L^2(q_T)}}{\|\|\phi\|\|_{L^2(q_T)}}$	$\|\|y_k\|\|_{L^2(Q_T)}$
109	0.145656	0.335074	0.500159
110	0.143198	0.327986	0.491718
111	0.145656	0.335074	0.500159
112	0.143198	0.327986	0.491718
113	0.145656	0.335074	0.500159
114	0.143198	0.327986	0.491718
115	0.145656	0.335074	0.500159
116	0.143198	0.327986	0.491718
117	0.145656	0.335074	0.500159
118	0.143198	0.327986	0.491718
119	0.145656	0.335074	0.500159
120	0.143198	0.327986	0.491718
121	0.145656	0.335074	0.500159
122	0.143198	0.327986	0.491718
123	0.145656	0.335074	0.500159
124	0.143198	0.327986	0.491718
125	0.145656	0.335074	0.500159
126	0.143198	0.327986	0.491718
127	0.145656	0.335074	0.500159
128	0.143198	0.327986	0.491718
129	0.145656	0.335074	0.500159
130	0.143198	0.327986	0.491718
131	0.145656	0.335074	0.500159
132	0.143198	0.327986	0.491718
133	0.145656	0.335074	0.500159
134	0.143198	0.327986	0.491718
135	0.145656	0.335074	0.500159
136	0.143198	0.327986	0.491718
137	0.145656	0.335074	0.500159
138	0.143198	0.327986	0.491718
139	0.145656	0.335074	0.500159
140	0.143198	0.327986	0.491718
141	0.145656	0.335074	0.500159
142	0.143198	0.327986	0.491718
143	0.145656	0.335074	0.500159
144	0.143198	0.327986	0.491718
145	0.145656	0.335074	0.500159
146	0.143198	0.327986	0.500159
147	0.145656	0.335074	0.500159
148	0.143198	0.327986	0.491718
149	0.145656	0.335074	0.500159
150	0.143198	0.327986	0.491718
151	0.145656	0.335074	0.500159
152	0.143198	0.327986	0.491718
153	0.145656	0.335074	0.500159
154	0.143198	0.327986	0.491718
155	0.145656	0.335074	0.500159
156	0.143198	0.327986	0.491718

iteration k	$\frac{\|\|y_k - y_{k-1}\|\|_{L^2(Q_T)}}{\|\|y_k\|\|_{L^2(Q_T)}}$	$\frac{\|\|\phi_k - \phi_{k-1}\|\|_{L^2(q_T)}}{\|\|\phi\|\|_{L^2(q_T)}}$	$\|\|y_k\|\|_{L^2(Q_T)}$
157	0.145656	0.335074	0.500159
158	0.143198	0.327986	0.491718
159	0.145656	0.335074	0.500159
160	0.143198	0.327986	0.491718
161	0.145656	0.335074	0.500159
162	0.143198	0.327986	0.491718
163	0.145656	0.335074	0.500159
164	0.143198	0.327986	0.491718
165	0.145656	0.335074	0.500159
166	0.143198	0.327986	0.491718
167	0.145656	0.335074	0.500159
168	0.143198	0.327986	0.491718
169	0.145656	0.335074	0.500159
170	0.143198	0.327986	0.491718
171	0.145656	0.335074	0.500159
172	0.143198	0.327986	0.491718
173	0.145656	0.335074	0.500159
174	0.143198	0.327986	0.491718
175	0.145656	0.335074	0.500159
176	0.143198	0.327986	0.491718
177	0.145656	0.335074	0.500159
178	0.143198	0.327986	0.491718
179	0.145656	0.335074	0.500159
180	0.143198	0.327986	0.491718
181	0.145656	0.335074	0.500159
182	0.143198	0.327986	0.491718
183	0.145656	0.335074	0.500159
184	0.143198	0.327986	0.491718
185	0.145656	0.335074	0.500159
186	0.143198	0.327986	0.491718
187	0.145656	0.335074	0.500159
188	0.143198	0.327986	0.491718
189	0.145656	0.335074	0.500159
190	0.143198	0.327986	0.491718
191	0.145656	0.335074	0.500159
192	0.143198	0.327986	0.491718
193	0.145656	0.335074	0.500159
194	0.143198	0.327986	0.491718
195	0.145656	0.335074	0.500159
196	0.143198	0.327986	0.491718
197	0.145656	0.335074	0.500159
198	0.143198	0.327986	0.491718
199	0.145656	0.335074	0.500159
200	0.143198	0.327986	0.491718

TABLE 6.2 – Norms of the difference between y_k, y_{k-1} the state and ϕ_k, ϕ_{k-1} the control defined by Newton algorithm

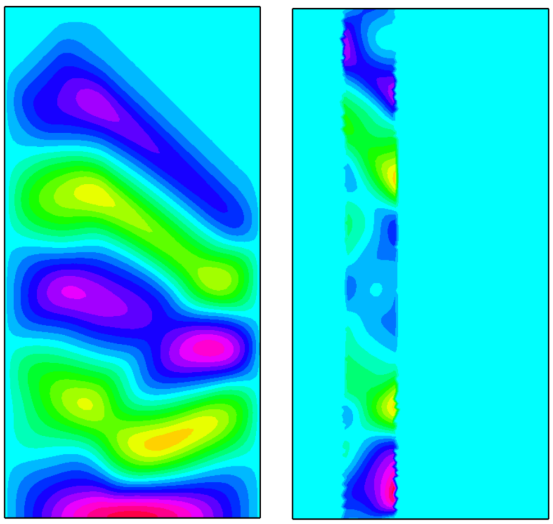

FIGURE 6.5 – $(y^0, y^1) = (sin(\pi x), 0)$, $f(s) = 300s.\log^2(1 + |s|)$.

Now we introduce the method of continuation, where we solve the problem for smaller $\lambda \in (0, 1)$ and then we increment it in order to get to $\lambda = 1$.

iteration k	$\frac{\|\|y_k - y_{k-1}\|\|_{L^2(Q_T)}}{\|\|y_k\|\|_{L^2(Q_T)}}$	$\frac{\|\|\phi_k - \phi_{k-1}\|\|_{L^2(q_T)}}{\|\|\phi\|\|_{L^2(q_T)}}$	$\|\|y_k\|\|_{L^2(Q_T)}$	λ
1	0.896025	3.0435	0.426799	0.7
2	1.73917	1.69089	0.790283	0.7
3	0.615372	1.41668	0.733764	0.7
4	0.671613	1.04192	0.698534	0.7
5	0.394005	0.838389	0.619956	0.7
6	0.184589	0.688638	0.567402	0.7
7	0.100993	0.239455	0.544706	0.7
8	0.0489771	0.117224	0.533206	0.7
9	0.0288513	0.0647382	0.527496	0.7
10	0.0198665	0.0439407	0.523823	0.7
11	0.0157729	0.034165	0.521162	0.7
12	0.0135636	0.0288155	0.518998	0.7
13	0.0123907	0.0257426	0.517101	0.7
14	0.0118836	0.0241115	0.515335	0.7
15	0.011905	0.0235719	0.513612	0.7
16	0.0124243	0.0239949	0.511861	0.7
17	0.0134936	0.0254002	0.510011	0.7
18	0.0152494	0.0279452	0.507985	0.7
19	0.0179378	0.0319471	0.50569	0.7
20	0.0219575	0.0379283	0.503006	0.7
21	0.0278987	0.0466572	0.499792	0.7
22	0.0364737	0.0590685	0.495905	0.7
23	0.0479916	0.0757224	0.491295	0.7
24	0.0608196	0.095295	0.486167	0.7
25	0.069976	0.113169	0.481065	0.7

iteration k	$\frac{\|\|y_k - y_{k-1}\|\|_{L^2(Q_T)}}{\|\|y_k\|\|_{L^2(Q_T)}}$	$\frac{\|\|\phi_k - \phi_{k-1}\|\|_{L^2(q_T)}}{\|\|\phi\|\|_{L^2(q_T)}}$	$\|\|y_k\|\|_{L^2(Q_T)}$	λ
26	0.0707852	0.124072	0.476661	0.7
27	0.0616595	0.120597	0.473499	0.7
28	0.0388674	0.0860938	0.471638	0.7
29	0.0143849	0.0360365	0.470874	0.7
30	0.0037357	0.00958624	0.470656	0.7
31	0.000724701	0.00217329	0.47058	0.7
32	0.000351193	0.000686804	0.470583	0.7
33	0.000219105	0.000476169	0.470565	0.7
34	0.000210941	0.000421957	0.470578	0.7
35	0.000187239	0.000380298	0.470566	0.7
36	0.000169688	0.00034322	0.470577	0.7
37	0.000153	0.000309728	0.470567	0.7
38	0.000138117	0.000279532	0.470576	0.7
39	0.000124637	0.000252259	0.470567	0.7
40	0.000112486	0.000227664	0.470575	0.7
41	0.000101512	0.000205454	0.470568	0.7
42	9.16143e-05	0.000185421	0.470574	0.7
1	0.18098	0.181901	0.464539	0.75
2	0.0759709	0.227063	0.472504	0.75
3	0.0680169	0.11094	0.471772	0.75
4	0.0359497	0.084848	0.475057	0.75
5	0.0417108	0.064489	0.477013	0.75
6	0.0360036	0.0620456	0.480081	0.75
7	0.0387982	0.0618147	0.483331	0.75
8	0.0391501	0.0648221	0.487142	0.75
9	0.0404156	0.068801	0.491437	0.75
10	0.0400137	0.0724227	0.49612	0.75
11	0.0367874	0.0721592	0.500808	0.75
12	0.0303729	0.0651293	0.504969	0.75
13	0.0225778	0.0522049	0.50823	0.75
14	0.0155288	0.0377579	0.510544	0.75
15	0.0101808	0.0254881	0.512082	0.75
16	0.00649441	0.0165351	0.513064	0.75
17	0.00408017	0.0104958	0.513679	0.75
18	0.00254175	0.00658211	0.514061	0.75
19	0.00157591	0.00409924	0.514295	0.75
20	0.000974499	0.0025425	0.51444	0.75
21	0.000601706	0.00157306	0.514529	0.75
22	0.000371214	0.000971798	0.514584	0.75
23	0.000228907	0.000599795	0.514617	0.75
24	0.000141117	0.000369982	0.514638	0.75
25	8.69836e-05	0.000228142	0.514651	0.75
1	0.100215	0.162605	0.506012	0.8
2	0.0187728	0.0830647	0.507881	0.8
3	0.00864753	0.0247315	0.508524	0.8
4	0.00315783	0.0102054	0.508858	0.8
5	0.00159047	0.00462716	0.509027	0.8
6	0.000726366	0.00219097	0.50911	0.8

iteration k	$\frac{\|\|y_k-y_{k-1}\|\|_{L^2(Q_T)}}{\|\|y_k\|\|_{L^2(Q_T)}}$	$\frac{\|\|\phi_k-\phi_{k-1}\|\|_{L^2(q_T)}}{\|\|\phi\|\|_{L^2(q_T)}}$	$\|\|y_k\|\|_{L^2(Q_T)}$	λ
7	0.00035824	0.00105385	0.509149	0.8
8	0.000171316	0.000509424	0.509169	0.8
9	8.35311e-05	0.00024684	0.509178	0.8
1	0.0979718	0.161526	0.500272	0.85
2	0.0181423	0.0882257	0.502622	0.85
3	0.00691926	0.0215447	0.503019	0.85
4	0.00258568	0.00794393	0.503182	0.85
5	0.00101904	0.00317551	0.503278	0.85
6	0.000428216	0.00127664	0.503307	0.85
7	0.000165974	0.000519152	0.503324	0.85
8	7.09714e-05	0.000211076	0.503329	0.85
1	0.0991574	0.162355	0.494782	0.9
2	0.0200037	0.100631	0.498095	0.9
3	0.00693674	0.0200925	0.498065	0.9
4	0.00184907	0.00644448	0.498289	0.9
5	0.000890419	0.00240051	0.498308	0.9
6	0.000305598	0.00100242	0.498342	0.9
7	0.000171695	0.000452225	0.498341	0.9
8	7.15939e-05	0.00021991	0.498347	0.9
1	0.103293	0.164253	0.490387	1
2	0.0290519	0.124755	0.494943	1
3	0.0188513	0.0437277	0.493873	1
4	0.0124105	0.0324097	0.494782	1
5	0.0102193	0.0251232	0.494183	1
6	0.00799359	0.0200772	0.494672	1
7	0.00640041	0.015934	0.494282	1
8	0.00510321	0.0127457	0.494588	1
9	0.00406524	0.0101405	0.49434	1
10	0.00324813	0.00810503	0.494534	1
11	0.00258724	0.0064554	0.494377	1
12	0.00206674	0.00515638	0.494501	1
13	0.00164702	0.0041095	0.494401	1
14	0.00131514	0.00328114	0.49448	1
15	0.00104847	0.002616	0.494417	1
16	0.000836944	0.00208811	0.494467	1
17	0.000667412	0.00166521	0.494427	1
18	0.000532657	0.00132894	0.494459	1
19	0.000424833	0.00105996	0.494433	1
20	0.000339012	0.000845816	0.494454	1
21	0.000270415	0.000674684	0.494437	1
22	0.00021577	0.000538338	0.49445	1
23	0.000172123	0.000429443	0.49444	1
24	0.000137333	0.000342641	0.494448	1
25	0.000109557	0.000273342	0.494441	1
26	8.74103e-05	0.000218086	0.494447	1

TABLE 6.3 – The solution (state) and the restriction of the minimal L^2 norm control on ω. in the case : $f(s) = 300s\log^2(1+|s|)$.

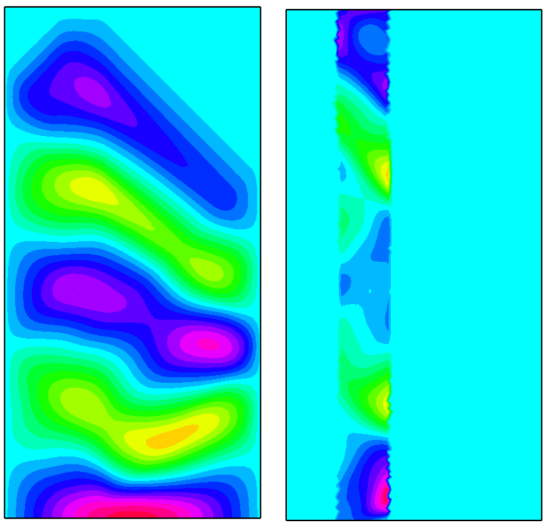

FIGURE 6.6 – The solution (state) and the restriction of the minimal L^2 norm control on ω. in the case : $f(s) = 300s\log^2(1+|s|)$.

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